Global, regional and national inequalities in neglected tropical disease and malaria burden in 204 countries and territories, 1990–2023, and the projected consequences of external financing cuts: an econometric modelling study

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Summary

Background

~~Neglected tropical diseases (NTDs) and malaria remain major contributors to global health loss, with substantial geographical inequalities. We assessed trends and the distribution of NTD and malaria burden across 204 countries and territories from 1990 to 2023 and examined associations with external health financing.~~Neglected tropical diseases (NTDs) and malaria are concentrated in countries that depend heavily on external health financing, which fell by about a fifth in 2025. We quantified how this burden is distributed, how socioeconomic inequality in it changed, and the health losses that financing cuts could cause.

Methods

~~We conducted a comparative ecological modelling study using Global Burden of Disease 2023 estimates of disability-adjusted life-years (DALYs) and age-standardised DALY rates from 1990 to 2023. We assessed temporal trends and geographical inequalities across countries and regions. Econometric analyses examined associations between external health financing and disease burden, using cross-sectional and longitudinal models with adjustment for sociodemographic development and other health-system and population covariates. Sensitivity analyses assessed alternative model specifications and exposure definitions.~~We analysed Global Burden of Disease (GBD) 2023 disability-adjusted life-years (DALYs) for 204 countries and territories, 1990–2023, by super-region, region and country. We measured spatial clustering, absolute and relative

inequality by Socio-demographic Index (SDI) and Theil decomposition. Panel models for 153 low- and middle-income countries, 2007–2022, related external financing share to DALY rates; two scenario models projected excess DALYs under 5–60% cuts.

Findings

~~In 2023, the 204 country estimates comprised 70.80 million combined NTD and malaria DALYs, including 51.56 million malaria DALYs and 19.24 million NTD DALYs. Sub-Saharan Africa accounted for 80.2% of combined DALYs despite representing 15.1% of the population; it accounted for 95.1% of malaria DALYs and 40.3% of NTD DALYs. Globally, the combined age-standardised DALY rate declined by 21.6% (95% uncertainty interval 7.8–36.3) between 2010 and 2023. In sub-Saharan Africa, the rate declined by 33.8% (21.1–49.9). Associations between external financing and disease burden varied by disease group and model specification.~~In 2023, NTDs and malaria caused 70.8 million DALYs (51.6 million from malaria). Sub-Saharan Africa had 15.1% of the world population but 80.2% of these DALYs; western and central sub-Saharan Africa alone had 63.1%. Age-standardised rates fell 21.6% (95% UI 7.8–36.3) from 2010. The absolute SDI gap narrowed (NTD slope index of inequality −1966 to −632 DALYs per 100 000, 1990–2023), but relative concentration in low-SDI countries persisted (malaria concentration index −0.80 in both years). Disability accounted for 62.4% of NTD DALYs. A 10-percentage-point higher external share was associated with a 13.4% (95% CI 6.4–20.0) lower NTD rate between countries but not within countries (−0.64%, −2.39 to 1.14). A 19–26% cut projected 0.41 million (95% UI 0.33–0.50) excess NTD and 1.85 million (1.51–2.21) excess malaria DALYs a year, over 90% in sub-Saharan Africa.

Interpretation

~~Despite declining age-standardised burden, NTDs and malaria remain unequally distributed, with sub-Saharan Africa carrying a disproportionate share of health loss. Disease-specific and geographically targeted strategies remain important. Associations with external financing should not be interpreted as causal effects.~~Falling averages have not moved burden away from the poorest, most donor-dependent countries, where projected losses also concentrate. Protecting service continuity there is a priority; the projections are conditional scenarios, not causal estimates.

Funding

This study was self-funded.

Research in context

Evidence before this study

We searched PubMed from inception to Oct 21, 2026, without language restrictions, for studies of NTD or malaria burden, inequality, spatial analysis, development assistance for health and funding cuts, and reviewed WHO and IHME reports; 76 of 704 unique records were eligible (appendix). GBD 2019 and 2021 analyses showed combined NTD and malaria DALYs falling by about 18% between 1990 and 2021, with burden concentrating in low-SDI countries and relative inequality widening as absolute inequality narrowed. None used GBD 2023, tested geographical clustering, or decomposed inequality between and within regions. Cross-country panels associated aid volume with lower malaria, NTD and child mortality

and projected 14–23 million additional deaths by 2030 from current cuts, but none separated between-country from within-country associations, examined dependence on external financing, or estimated NTD DALYs under the cuts.

Added value of this study

Using GBD 2023, we show that 63.1% of global NTD and malaria DALYs occur in western and central sub-Saharan Africa, that burden hot spots were unchanged between 2010 and 2023, and that the absolute socioeconomic gap narrowed while relative concentration in the poorest countries did not. Most NTD burden is disability, which mortality-based evaluations omit. Thirty-four countries combining high NTD burden and high donor dependence hold over half of malaria DALYs. The association of external financing with NTD burden seen between countries was absent within countries over time. We provide country-level estimates of NTD and malaria DALYs at risk under specified cuts.

Implications of all the available evidence

The 2025 reductions in development assistance fall where NTD and malaria burden and financing dependence are greatest, which supports protecting service continuity in those countries. Aggregate financing shares should not be read as disease-specific effects; estimating the consequences of the cuts needs programme disbursement data linked to coverage and outcomes.

Introduction

~~Control of neglected tropical diseases (NTDs) and malaria is central to Sustainable Development Goal 3.3 and international disease-elimination commitments, but progress remains incomplete (World Health Organization [WHO], 2020, 2025b). In 2023, 1.495 billion people required NTD interventions, despite a 32% reduction since 2010; malaria caused an estimated 282 million cases and 610,000 deaths in 2024 (WHO, 2025a, 2025b). These indicators capture different aspects of the continuing burden, while disability-adjusted life-years (DALYs) combine premature mortality and non-fatal health loss within a common measure (GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025). Declining global burden has nevertheless coexisted with persistent geographical and socioeconomic disparities (He et al., 2025; Lin et al., 2025).~~NTDs and malaria remain central to Sustainable Development Goal 3.3. In 2023, 1.495 billion people required NTD interventions, and in 2024 malaria caused an estimated 282 million cases and 610 000 deaths (World Health Organization [WHO], 2025a, 2025b). Combined DALYs fell by about 18% between 1990 and 2021, to 71.6 million (Zheng et al., 2025), and successive GBD rounds show burden concentrating in low-SDI countries and western sub-Saharan Africa (GBD 2023 Causes of Death Collaborators, 2025; Lin et al., 2022; Murray et al., 2014). Between-country inequality narrowed in absolute terms but widened in relative terms (Emadi et al., 2021; Feng et al., 2025; He et al., 2025; Lin et al., 2025; Stolk et al., 2016), and NTD treatment coverage remains uneven within countries (Khaki et al., 2025; Lo et al., 2019). None of these studies used GBD 2023, tested whether burden clusters geographically, or decomposed inequality between and within regions.

~~Analyses of GBD 2021 data show that NTD and malaria health losses remain concentrated in countries with lower sociodemographic development, particularly in sub-Saharan Africa (He et al., 2025; Lin et al., 2025). Lin et al. (2025) found reductions in absolute inequality without comparable improvement in relative inequality, indicating that falling burden does not necessarily narrow proportional differences between countries. Environmental exposure and material deprivation also operate together: a study across 21 African countries associated improved housing with lower odds of childhood malaria infection after adjustment for household wealth and malaria interventions (Tusting et al., 2017). For NTDs, access to~~

~~water, sanitation and sustained treatment remains integral to control and elimination strategies (WHO, 2020). These findings support examining the distribution of progress within and between regions alongside global trends (He et al., 2025; Lin et al., 2025).~~External financing underpinned much of this progress. Cross-national panels associate development assistance for health with lower malaria and child mortality (Akachi & Atun, 2011; Bendavid & Bhattacharya, 2014; Hsiao & Emdin, 2014; Mishra & Newhouse, 2009; Yan et al., 2015), and evaluations attribute large gains to the US President's Malaria Initiative and insecticide-treated nets (Bhatt et al., 2015; Jakubowski et al., 2017). Development assistance for health fell by an estimated 22% between 2024 and 2025 (Apeagyei et al., 2025). Recent studies project 14–23 million additional deaths by 2030 from USAID and broader aid cuts (Cavalcanti et al., 2025; A. F. da Silva et al., 2026), and models address HIV, tuberculosis and malaria (Hallett et al., 2025; Stover et al., 2025; Symons et al., 2025). These analyses model aid volume and mortality. Yet most NTD burden is non-fatal, and because aid flows towards high-burden countries (Namubiru, 2025; Sempungu et al., 2023), between-country associations can differ from within-country change. WHO already reports disruption of NTD programmes (World Health Organization Regional Office for Africa, 2025). No study has estimated NTD DALYs at risk from the cuts.

~~External financing has supported the expansion of disease-control services, with a difference-in-differences evaluation associating the US President's Malaria Initiative (PMI) with a 16% reduction in annual under-five mortality risk in recipient countries (Jakubowski et al., 2017). Financial disruption now threatens programme delivery: a 2025 WHO regional assessment, comprising 25 responses from 18 countries, documented implementation disruption and anticipated postponement, modification or cancellation of NTD treatment and surveillance activities if funding gaps persisted (World Health Organization Regional Office for Africa, 2025). Apeagyei et al. (2025) projected development assistance for health to decline from US$49.6 billion in 2024 to US$38.4 billion in 2025, approximately 22%. However, global assistance flows and external financing as a share of national current health expenditure measure different quantities; that projected decline does not establish an equivalent reduction in each country's disease-programme budget (Apeagyei et al., 2025; World Bank, n.d.).~~

~~Recent studies already estimate the consequences of losing programme support: Symons et al. (2025) projected that planned PMI activities would avert 13.6 million malaria cases and 104,000 deaths in 2025 compared with withdrawal without mitigation. For NTDs, transmission models of pandemic-related interruptions predicted that a one-year interruption could delay elimination targets by two to three years for several diseases in high-prevalence settings, demonstrating consequences that depend on transmission and programme conditions (Borlase et al., 2022). A subsequent analysis linked assistance to mortality in 93 low- and middle-income countries and projected deaths under alternative defunding scenarios (A. F. da Silva et al., 2026). These studies address programme withdrawal and mortality, complementing research on burden inequalities; together, they motivate examining how DALY burden, geographical inequalities and national financing dependence coincide (A. F. da Silva et al., 2026; Lin et al., 2025; Symons et al., 2025). Such projections require explicit assumptions, since historical financing–burden associations alone do not identify causal responses to future cuts (Imai & Kim, 2021).~~

~~We examine NTD and malaria burden across 204 countries and territories, using GBD 2023 estimates and World Bank indicators, with super-regional trends extending from 1990 to 2023 and an annual country panel covering 2005–2023 (GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025; World Bank, n.d.). We assess spatial clustering, socioeconomic inequality and disparities within and between super-regions, then examine where high burden coincides with external financing dependence. We distinguish between-country patterns from within-country financing–burden associations and estimate conditional health losses under alternative financing-reduction scenarios. The aim is to inform financing~~

priorities and continuity of NTD and malaria services where substantial disease burden and financing vulnerability overlap.We analysed NTD and malaria burden in 204 countries and territories using GBD 2023, at super-region, regional and national level. We quantified spatial clustering, absolute and relative socioeconomic inequality from 1990 to 2023, and the fatal and non-fatal composition of burden; compared between-country and within-country associations of external financing dependence with burden; and projected excess DALYs under specified financing cuts.

Methods

Study designStudy design and data

We conducted an ecological panel study of 204 countries and territories in seven Global Burden of Disease (GBD) super-regions, covering 2005–2023. We described the burden of neglected tropical diseases (NTDs) and malaria, assessed spatial clustering and socioeconomic inequality, examined associations with external health financing, and modelled potential health losses under reductions in development assistance for health (DAH). The unit of analysis was the country-year; estimates therefore describe population-level associations rather than individual effects (Morgenstern, 1995). Reporting followed the Guidelines for Accurate and Transparent Health Estimates Reporting (GATHER) and relevant STROBE recommendations (Stevens et al., 2016; von Elm et al., 2007). The GATHER checklist is provided in the appendix.We conducted an ecological study of 204 countries and territories in seven GBD super-regions and 21 regions, reported according to GATHER and STROBE (appendix section 1) (Stevens et al., 2016; von Elm et al., 2007). We linked GBD 2023 estimates of DALYs, years lived with disability (YLDs) and years of life lost (YLLs), World Bank World Development Indicators (WDI), WHO indicators from the UN SDG database, IHME Financing Global Health estimates and Natural Earth boundaries (Table S1; Figure 1) (Apeagyei et al., 2025; GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025; World Bank, n.d.). Outcomes were age-standardised DALY rates per 100 000 for NTDs excluding malaria (the GBD NTD-and-malaria aggregate minus malaria; 19 causes) and for malaria; counts were all-age. Percentage changes with uncertainty intervals (UIs) were taken from GBD. GBD does not publish changes for NTDs excluding malaria or country-level changes for malaria alone; we computed these from point estimates, without intervals.

Data sources and linkage

We linked five public sources: GBD 2023, World Bank World Development Indicators (WDI), World Health Organization (WHO) indicators disseminated through the UN Sustainable Development Goal (SDG) database, Institute for Health Metrics and Evaluation (IHME) Financing Global Health estimates, and Natural Earth boundaries (Apeagyei et al., 2025; GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025; United Nations Statistics Division, n.d.; World Bank, n.d.; World Health Organization, n.d.). Records were harmonised using ISO 3166-1 alpha-3 codes, GBD location identifiers and the GBD super-region hierarchy. Table S1 summarises source coverage; Figure 1 shows the analytical workflow.

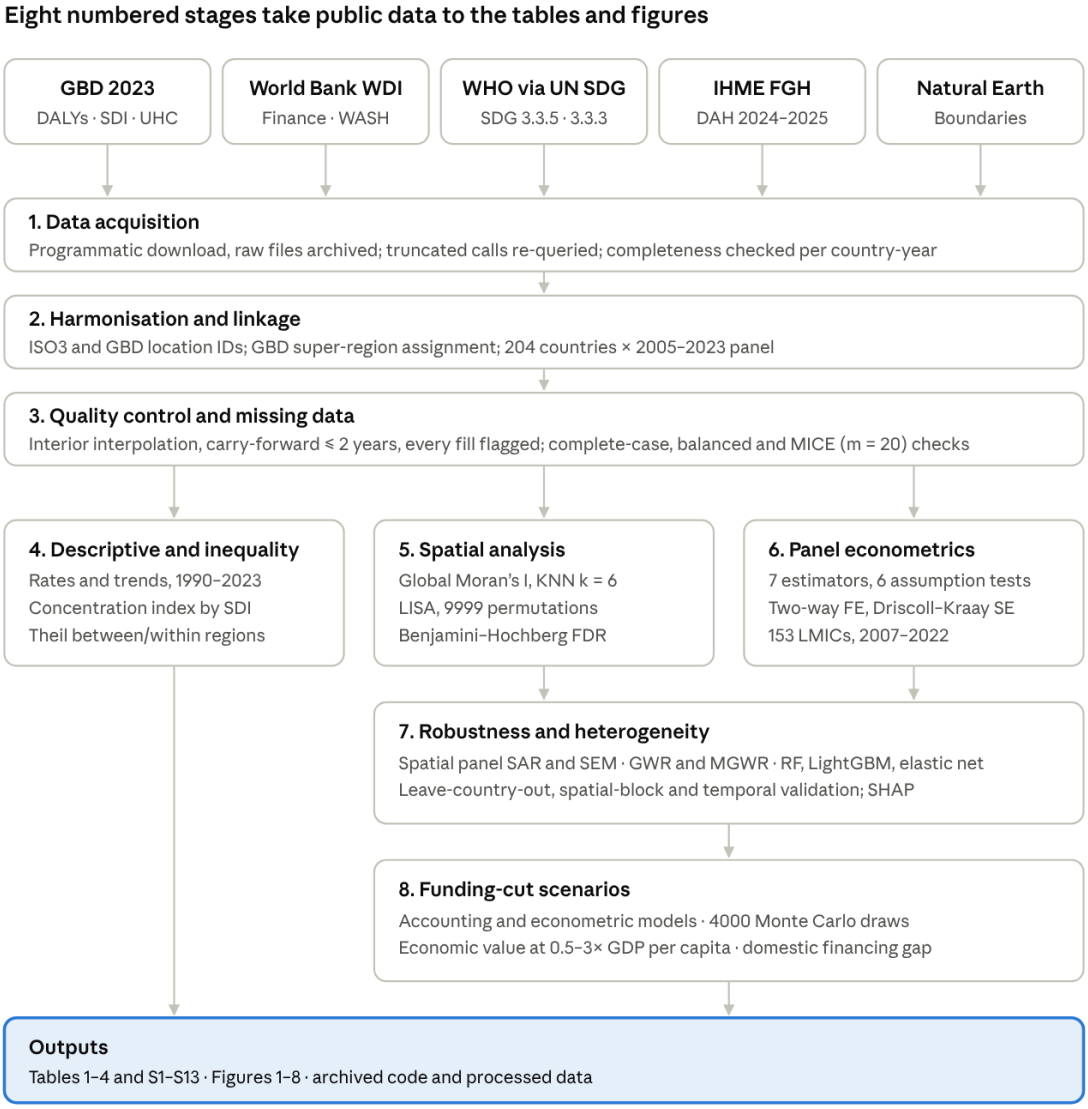


Figure 1: Analytical workflow. Data acquisition, linkage and quality checks produced a country-year panel for descriptive, inequality, spatial and panel analyses. Panel estimates informed sensitivity analyses and funding-cut scenarios

~~Study populations~~

~~Descriptive analyses covered 204 GBD locations. Spatial analyses included 203 locations with Natural Earth polygons, excluding Tokelau. Financing analyses began with 168 locations outside the GBD high-income super-region; this geographical classification is not equivalent to World Bank low- and middle-income status. After covariate exclusions, the NTD panel included 153 locations. Malaria models additionally required a maximum age-standardised malaria DALY rate above 10 per 100 000 during 2005–2023: 78 locations met this threshold and 77 entered the complete-case panel. Inequality analyses required population data; the original Theil analysis also excluded zero-rate locations.~~

~~Burden outcomes~~

~~GBD 2023 supplied annual all-age DALY counts and age-standardised DALY rates for the combined NTD-and-malaria category and malaria during 2005–2023. NTD totals excluding malaria were obtained by subtraction. The 2023 cross-section additionally contained 19 individual NTD components, selected deaths and prevalence estimates, and published uncertainty intervals (UIs). The 19-component DALY counts reproduced the 2023 NTD total. Historical country-level extracts covered selected years, whereas super-regional trends extended from 1990 to 2023. Age-standardised rates were expressed per 100 000 population (GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025). Percentage changes and their UIs were imported from GBD rather than calculated from endpoint bounds.~~

~~Financing exposures~~Financing exposure and covariates

~~The primary exposure was external health expenditure as a share of current health expenditure (CHE), scaled per 10 percentage points. The secondary exposure was log(1 + external health expenditure per capita). WDI health-financing series derive from the WHO Global Health Expenditure Database (GHED), which follows the System of Health Accounts 2011 (World Bank, n.d.; World Health Organization, n.d.). External expenditure includes bilateral, multilateral and philanthropic financing originating outside the country. Both exposures were lagged by 2 years to allow for programme implementation and reduce contemporaneous feedback; lagging alone does not resolve endogeneity (Bellemare et al., 2017; Bendavid & Bhattacharya, 2014). Monetary values were expressed in current US dollars.~~The primary exposure was external health expenditure as a share of current health expenditure, per 10 percentage points, which measures dependence; the secondary exposure was log(1 + external health expenditure per capita), which measures volume (World Bank, n.d.). Both were lagged by 2 years. Models adjusted for log domestic government health expenditure per capita (lagged 2 years), log lag-distributed income per capita, universal health coverage (UHC) effective coverage, and basic sanitation and drinking-water coverage, selected a priori (Bendavid & Bhattacharya, 2014; Mishra & Newhouse, 2009; WHO, 2020).

~~Additional sources and data quality~~

~~WHO-reported indicators provided outcomes not obtained from GBD: the number of people requiring NTD interventions (SDG 3.3.5) and malaria incidence per 1000 people at risk (SDG 3.3.3) (United Nations Statistics Division, n.d.; World Health Organization, 2020). These were retrieved through the UN SDG database after the WHO API returned server errors. IHME Financing Global Health estimates for 1990–2030 were used to calibrate the funding-cut scenarios (Apeagyei et al., 2025). Estimated DAH fell from US$49.6 billion in 2024 to a preliminary US$38.4 billion in 2025, approximately 22%. Spatial analyses used Natural Earth administrative level 0 polygons at 1:50 million scale (Natural Earth, n.d.). After projection to Robinson coordinates (ESRI:54030), the scripts generated an interior representative point for each country polygon. Euclidean distances between these points defined the original spatial weights. GBD extracts were retained as raw CSV and JSON files; WDI and SDG series were retrieved through public APIs. Truncated requests were split into shorter year windows, and country-year coverage was checked after assembly. The GBD outcome panel contained 3876 observations (204 locations × 19 years), with no missing outcomes. Raw data, processed datasets and scripts are archived with the paper. Natural Earth records with missing ISO codes were matched using administrative codes. The supplied WDI extract covered 2005–2024; the panel retained 2005–2023. Table S11 lists indicator codes, definitions, transformations and lags.~~

Missing data

~~GBD outcomes and the GBD covariates including Socio-demographic Index (SDI), universal health coverage (UHC) effective-coverage index and lag-distributed income per capita (LDI) were complete. Missingness affected WDI series. In the modelling frame of 2688 country-years (168 locations, 2007–2022), missing values accounted for 15.1% of lagged external expenditure share, 8.2% of lagged domestic government health expenditure, 4.7% of sanitation coverage and 4.3% of drinking-water coverage (Table S10a).~~GBD outcomes and covariates were complete. Gaps in WDI series (15.1% for lagged external share) were linearly interpolated within countries, with last observations carried forward for up to 2 years; remaining incomplete records were excluded. The analysis samples comprised 153 countries (2287 country-years) for NTDs and 77 malaria-endemic countries (1187 country-years) for malaria, 2007–2022. Complete-case, balanced-panel and multiple-imputation analyses assessed sensitivity (Tables S10a–b).

~~Fourteen locations had no usable lagged external expenditure share in the modelling window: American Samoa, Bermuda, Cook Islands, Guam, Montenegro, Northern Mariana Islands, Niue, Puerto Rico, the Democratic People's Republic of Korea, Slovakia, Tokelau, Türkiye, Taiwan and the US Virgin Islands. Nicaragua also lacked sanitation and drinking-water observations. These exclusions reduced the eligible NTD sample from 168 to 153 locations.~~

~~We assessed associations between incomplete records and log NTD DALY rate, SDI, year and super-region using logistic regression with country-clustered standard errors. These diagnostics described observed missingness patterns but could not establish whether data were missing at random.~~

~~Within each country, interior WDI gaps were linearly interpolated without a maximum gap length; final observations were carried forward for at most 2 years. We did not extrapolate backwards or impute outcomes. Remaining incomplete records were excluded from each model. We reconstructed a model-specific flag from the observed input series, using exposure values 2 years before the outcome and contemporaneous sanitation and drinking-water values. This identified filled model inputs in 62 of 2287 NTD observations (2.7%) and 14 of 1187 malaria observations (1.2%).~~

~~Missing-data sensitivity analyses used observed model inputs only, restricted the sample to countries with complete observations throughout 2007–2022, and used multiple imputation (Honaker & King, 2010; Rubin, 1987; Sterne et al., 2009; van Buuren & Groothuis-Oudshoorn, 2011; White et al., 2011). Excluding observations with the corrected model-specific filling flag produced the same sample as requiring observed inputs, so these checks are not independent robustness tests. The original imputation model used four WDI variables, LDI, UHC, SDI, year, the observed outcome and country means of observed covariates. Outcomes were predictors only.~~

~~The supplied multiple-imputation implementation used Bayesian ridge regression with posterior sampling, 15 iterations and 20 datasets. Missing country-level auxiliary means were replaced with the corresponding across-country median. It imposed no explicit bounds on percentage variables and recorded no convergence diagnostics. Estimates were pooled using Rubin's rules (Rubin, 1987). Table S10b retains these original estimates as provisional pending imputation diagnostics; its variance summary is the proportion of total variance attributable to the between-imputation component, not a finite-sample fraction-of-missing-information estimate.~~

~~Cross-sectional exposures used the latest available observation during 2019–2023. Complete records were available for 169 of 204 locations globally; 145 remained after excluding the GBD high-income super-region. Table S9 describes the global complete-case comparison. The WHO NTD intervention-need proportion divided the 2023 count by the latest available WDI population value, so its denominator was not necessarily from 2023.~~

~~Scenario analyses used 2022 population where 2023 values were unavailable. Missing GDP per capita for South Sudan, Yemen and Eritrea was replaced by the study-sample median for monetary valuation only, not for DALY projections.~~

~~Model specification~~

~~NTD rates were natural-log transformed; malaria rates used log (1 + rate) to accommodate values approaching zero. NTD and malaria models were fitted separately. Covariates were selected a priori from aid–health studies and the WHO NTD road map (Bendavid & Bhattacharya, 2014; Mishra & Newhouse, 2009; World Health Organization, 2020). Models adjusted for log domestic government health expenditure per capita (lagged 2 years), log LDI, UHC effective coverage per 10 points, and basic sanitation and drinking-water coverage per 10 percentage points (GBD 2019 Universal Health Coverage Collaborators, 2020). SDI was excluded from the primary models because it incorporates income; it was used for inequality ranking and sensitivity analyses. Collinearity was assessed using variance inflation factors on within-transformed covariates; maximum values were 6.1 for NTDs and 6.4 for malaria. These were interpreted as diagnostics rather than against an absolute threshold (O'Brien, 2007).~~

Descriptive and inequality analyses

~~We summarised 2023 all-age DALY counts, shares of global burden and age-standardised rates by super-region, and changes in age-standardised rates from 2010 to 2023 (Table 1). Cause-specific estimates are provided in Table S2. Maps, heat maps and trend plots describe geographical and temporal patterns (Figure 2, Figure 3 & Figure 4). Socioeconomic inequality was measured using a population-weighted concentration index, ranking countries from lowest to highest 2023 SDI (Kakwani et al., 1997; O'Donnell et al., 2008; Wagstaff et al., 1991). The index equals twice the weighted covariance of the age-standardised rate and fractional socioeconomic rank, divided by the weighted mean rate. Negative values indicate concentration in lower-SDI countries. We obtained 95% confidence intervals (CIs) from 2000 country-level bootstrap resamples using the percentile method.~~We summarised burden by super-region, region and country, ranked countries by count and rate, and identified countries whose rates rose between 2010 and 2023. Socioeconomic inequality was measured in 1990, 2010 and 2023, ranking countries by SDI of the same year and weighting by population. Relative inequality was measured by the concentration index and absolute inequality by the slope index of inequality (SII), the modelled difference in rate between the highest-SDI and lowest-SDI ends of the distribution (Kakwani et al., 1997; Mackenbach & Kunst, 1997; Wagstaff et al., 1991). Country bootstrap resampling (2000 replicates) gave 95% CIs, with paired resampling for changes over time. Theil T was decomposed into components between super-regions, between regions within super-regions and within regions (Shorrocks, 1980; Theil, 1967). GBD rates were compared with WHO-reported people requiring NTD interventions and malaria incidence by Spearman correlation.

~~We calculated the population-weighted Theil T index and decomposed it into between-super-region and within-super-region components (Table 2; Figure 5) (Shorrocks, 1980; Theil, 1967). The original calculation required positive rates and available population, yielding 199 locations for NTDs and 86 for malaria. A sensitivity analysis retained zero malaria rates among all 199 locations with population data, assigning zero to the limiting term 0 log (0).~~

Spatial autocorrelation

~~Spatial dependence was assessed before regression modelling (Anselin, 1988). We constructed row-standardised weights from the six nearest neighbours of each projected country representative point. This retained island states in the spatial analysis. Sensitivity analyses used four or eight neighbours and a 2500~~

~~km inverse-distance band (Table S7) (Anselin, 1988; Getis & Aldstadt, 2004). Global Moran's I assessed clustering in log (1 + age-standardised DALY rate), with 9999 permutations for inference (Moran, 1950). We reported permutation p values and z scores. Analyses were repeated within the sample outside the GBD high-income super-region and, for malaria, within endemic locations. Local Moran statistics identified high–high and low–low clusters and high–low or low–high outliers (Anselin, 1995). Multiple testing across 203 locations was addressed using the Benjamini–Hochberg false-discovery-rate procedure at q < .05 (Benjamini & Hochberg, 1995). Cluster maps are shown in Figure 5.~~Global Moran's I and local indicators of spatial association were computed on log(1 + rate) with six-nearest-neighbour weights and 9999 permutations; local clusters were reported at a Benjamini–Hochberg false-discovery rate below 5% (Anselin, 1995; Benjamini & Hochberg, 1995; Moran, 1950). Analyses were repeated for 1990 and 2010 and with alternative weights (Table S7).

~~Panel econometric models~~Financing models

~~The primary analysis used within-country variation to account for time-invariant country characteristics (Wooldridge, 2010). Cross-sectional ordinary least squares (OLS) with super-region fixed effects and a maximum-likelihood spatial error model provided between-country comparisons (Bendavid & Bhattacharya, 2014; Cavalcanti et al., 2025; Mishra & Newhouse, 2009). Cross-sectional models additionally adjusted for rural population share. NTD panel models included 153 countries and 2287 country-years during 2007–2022; malaria models included 77 countries and 1187 country-years. The preferred specification included country and year fixed effects, the 2-year-lagged financing exposure and the covariates listed above. We retained 2007–2022 as the original primary window and examined extension through 2023 in a post-review sensitivity analysis. Additional checks omitted UHC and excluded observations without matching UHC or LDI records in the supplied raw archive (Table S15). We compared pooled OLS, random effects, two-way fixed effects with country-clustered standard errors, correlated random effects (Mundlak), first differences, and fixed effects with country-specific linear trends (Bell & Jones, 2015; Mundlak, 1978; Wooldridge, 2010). Mundlak models included country means of time-varying regressors. Country-specific trends were removed from outcomes and regressors before estimation. Each estimator was fitted using both financing exposures.~~The primary analysis estimated within-country associations with two-way (country and year) fixed effects and Driscoll–Kraay standard errors, which allow for the heteroskedasticity, serial correlation and cross-sectional dependence found in diagnostic tests (Table S3) (Driscoll & Kraay, 1998; Hoechle, 2007; Wooldridge, 2010). NTD rates were log-transformed and malaria rates log(1 + rate)-transformed. We compared this model with a 2023 cross-section of 145 countries, pooled, random-effects, first-difference and country-trend models, spatial panel models, and a correlated random-effects (Mundlak) model that separates between-country from within-country associations in one sample (Bell & Jones, 2015; Mundlak, 1978). Estimates are percentage differences per 10-percentage-point higher external share, interpreted as conditional associations rather than causal effects (Imai & Kim, 2021). Malaria mortality was an alternative outcome. Geographically weighted regression and machine-learning prediction, which assessed spatial heterogeneity and predictive performance, are described in the appendix.

~~Diagnostics comprised the Breusch–Pagan test for country-level heterogeneity, Hausman and cluster-robust Mundlak tests, Pesaran's test for cross-sectional dependence, the Wooldridge test for serial correlation, and a modified Wald test for groupwise heteroskedasticity (Table S3) (Bell & Jones, 2015; Breusch & Pagan, 1980; Drukker, 2003; Greene, 2018; Hausman, 1978; Pesaran, 2021). The preferred two-way fixed-effects model used Driscoll–Kraay standard errors with a Bartlett kernel and bandwidth of 3 to address heteroskedasticity, serial correlation and cross-sectional dependence (Driscoll & Kraay, 1998; Hoechle, 2007). Sensitivity analyses used bandwidths of 1, 2 and 4 and country-clustered standard errors. These checks assess inference sensitivity; they do not remove the limitations of a 16-year panel.~~

For NTDs, coefficients and confidence limits were transformed as $100 \times [\exp(\beta) - 1]$ to express percentage differences in DALY rates. For malaria, this transformation applies to rate + 1 and approximates a percentage difference in the rate when rates are large relative to 1. For the per-capita exposure, multiplying the coefficient by log (2) estimates a doubling of expenditure plus US$1, not a doubling of expenditure itself. All coefficients were interpreted as conditional associations. Fixed effects do not remove time-varying confounding, and lagging does not establish causal identification (Bellemare et al., 2017; Imai & Kim, 2021). We assessed consistency across estimators rather than relying on statistical significance alone.

Additional spatial and predictive analyses

Spatial panel models

Models used row-standardised six-nearest-neighbour weights and maximum likelihood with the Lee–Yu fixed-effects transformation (Lee & Yu, 2010). Year effects were removed by cross-country demeaning. Analyses were restricted to balanced 16-year panels of 130 countries for NTDs and 71 for malaria. The intended SAR summaries are direct, indirect and total associations derived from the spatial multiplier $(I - \rho W)^{-1}$ (LeSage & Pace, 2009) (Table S12).

Geographically weighted regression

GWR and MGWR examined the 2023 NTD association in 145 locations outside the GBD high-income super-region with complete inputs (Brunsdon et al., 1996; Fotheringham et al., 2017; Oshan et al., 2019). Financing exposure was mean external expenditure share during 2019–2022. The log NTD-DALY rate and all four predictors (external share, UHC, log LDI and sanitation coverage) were standardised to mean 0 and SD 1. Models used an adaptive bisquare kernel. Bandwidths were selected by minimising the corrected Akaike information criterion (AICc), using golden-section search for GWR and back-fitting for MGWR (Hurvich et al., 1998). MGWR estimated a separate bandwidth for each covariate. Local inference used the MGWR multiple-testing adjustment based on covariate-specific effective numbers of parameters (A. R. da Silva & Fotheringham, 2016). Fit was compared with global OLS using AICc and R^2 (Tables S6a–b; Figure 7C).

Predictive modelling

We compared elastic net, random forest and LightGBM for prediction of log NTD-DALY rates during 2007–2023, using validation schemes that reflected the panel and spatial structure (Bergmeir & Benítez, 2012; Breiman, 2001; Ke et al., 2017; Ploton et al., 2020; Roberts et al., 2017; Zou & Hastie, 2005). Elastic-net penalty strength and mixing proportion (0.1, 0.5 or 0.9) were selected by internal five-fold cross-validation. Random forest used 500 trees, a minimum leaf size of 5 and 50% of features at each split. LightGBM used 600 trees, learning rate 0.03, 15 leaves, and row and column subsampling of 0.8. Tree-model hyperparameters were fixed before test-set evaluation. In the supplied elastic-net implementation, scaling preceded internal cross-validation and ordinary internal folds did not preserve the grouped or temporal validation structure. These results require rerunning with preprocessing and tuning inside structure-appropriate training folds. All models used 13 features: both financing exposures, the five panel covariates, rural population share, out-of-pocket expenditure share, SDI, year, and two projected coordinate variables. Validation comprised random five-fold splits, grouped five-fold splits by country, leave-one-super-region-out validation, and temporal validation (training: 2007–2017; testing: 2018–2023). Random splits were included to illustrate potentially optimistic performance when countries appear in both training and test sets. Temporal validation included a linear benchmark with country fixed effects and a linear time trend, allowing extrapolation beyond the training years. Test-period covariates

~~were observed, so temporal results describe conditional predictions rather than fully prospective forecasts. Numerical comparisons remain provisional pending consistent test samples and corrected nested tuning. Performance was assessed on the log scale using root-mean-square error (RMSE), mean absolute error and R^2 (Table S5; Figure 7B). Mean absolute Shapley additive explanations (SHAP) values summarised feature contributions to LightGBM predictions in the full dataset (Lundberg & Lee, 2017). These were interpreted as model attribution, not causal effects.~~

~~Scenario modelling of development assistance cuts~~Scenarios of financing cuts

~~The original analysis used two conditional scenarios for annual health losses under external-financing reductions. Neither identifies a causal effect. Scenario estimates retained below, in Table 4 and in Figure 8 are provisional: the available data do not support the stated 19-cause NTD accounting calculation, and the malaria econometric conversion requires correction. The scenarios reduced each country's external expenditure by 5–15% (mild), 19–26% (reference) or 40–60% (severe). The reference range encompassed IHME's estimated 22% decline in global DAH between 2024 and 2025; the severe scenario represented larger reductions (Apeagyei et al., 2025; Cavalcanti et al., 2025). The same proportional cut was applied to 153 locations with financing data, using the latest available 2019–2023 financing baseline and 2023 burden and population.~~Two models projected annual excess DALYs if each country's external health expenditure fell by 5–15% (mild), 19–26% (reference, bracketing the 22% fall in development assistance for health in 2025) or 40–60% (severe) in 153 low- and middle-income countries (Apeagyei et al., 2025). The accounting model assumed that the fall in each country's all-age DALY rate between 2010 and 2023, summed over the 19 NTD causes or for malaria, was attributable to external financing in proportion to its share of health expenditure and would reverse in proportion to the cut. The econometric model applied the preferred fixed-effects coefficient to the cut-induced fall in external share; for malaria, the change was computed on the (1 + rate) scale and converted to the implied change in rate. Uncertainty was propagated through 4000 Monte Carlo draws of GBD estimates, cut size and coefficients (Briggs et al., 2012). DALYs were valued at one times GDP per capita (0.5 and 3 times in sensitivity analyses), and the rise in domestic government health expenditure needed to replace lost external funds was calculated by super-region.

~~The accounting script summed cause-specific changes in all-age DALYs per person between 2010 and 2023, truncated each country's total gain at zero, and multiplied it by 2023 population, external expenditure share and the funding-cut proportion. This assumes that gains attributable to external financing equal its share of health spending and reverse proportionately after a cut. Five 2010 NTD cause baselines were absent from the supplied extracts: other NTDs, yellow fever, Ebola, Guinea worm disease and Zika. The script replaced these missing values with near-zero counts. That substitution is unsupported, and the resulting NTD projections cannot be treated as validated estimates. The econometric script recalculated external expenditure share as $s(1 - c)/(1 - sc)$, where s is the baseline share expressed as a fraction and c is the proportional cut. It multiplied the share change in units of 10 percentage points by the preferred panel coefficient and applied $\exp(\beta\Delta x) - 1$ to baseline all-age DALYs. For NTDs, this additionally assumes that an age-standardised association transfers to all-age counts. For malaria, the outcome is log(1 + rate), so this proportional count conversion is only an approximation and must be replaced by an explicitly defined rate-scale calculation.~~

~~Uncertainty was propagated through 4000 Monte Carlo draws (Briggs et al., 2012). The scenario code treated each published DALY point estimate as a log-normal median and set the log-scale standard deviation to [log(upper UI) − log(lower UI)]/3.92. Simulated draws were independent across countries and causes, with shared standard-normal draws between 2010 and 2023 in the primary analysis and independent year draws in sensitivity analysis. These are approximations from published summaries, not~~

~~original GBD posterior draws, and generally do not reproduce both published bounds exactly. Each simulation sampled a funding cut uniformly within the scenario range and applied it to all countries. For the econometric model, coefficients were sampled from a normal distribution using the preferred estimate and Driscoll–Kraay standard error. Country estimates were summed within each draw to obtain super-regional and overall projections. We reported medians and 2.5th–97.5th percentiles; these intervals reflect the specified simulation assumptions. Projected DALYs were valued at one time each country's GDP per capita in current US dollars, with 0.5 and 3 times GDP per capita examined in sensitivity analyses. These multiples provide an illustrative monetary valuation, not a cost-effectiveness threshold or an estimate of fiscal losses (Bertram et al., 2016; Fitzpatrick et al., 2017; Marseille et al., 2015). The domestic financing gap was calculated at each scenario midpoint as external expenditure per capita × population × proportional cut, summed by super-region. This was expressed as a percentage of domestic government health expenditure to describe the increase needed to replace lost external funds (Table S14).~~

~~Sensitivity analyses~~

~~Sensitivity analyses examined alternative spatial weights and samples; financing exposure definitions; SDI substitution and WHO-reported outcomes; panel estimators and country trends; spatial models; missing-data strategies; and included versus excluded locations. Scenario analyses varied temporal dependence of burden draws, attribution of gains to external financing, and monetary valuation (Tables S7, S10b, S12 and S13).~~

Statistical software and reproducibility

~~Analyses used Python: pandas and NumPy for data management; statsmodels and linearmodels for regression and diagnostics; PySAL packages for spatial analyses; scikit-learn for imputation, elastic net, random forest and validation; LightGBM and SHAP for predictive modelling; and GeoPandas and matplotlib for maps and figures (Oshan et al., 2019; Rey & Anselin, 2007). The reproducible code cand be found on this GitHub repository. Random processes used fixed seeds (scenario seed: 20261007). Tests were two-sided at α = .05, with emphasis on estimates and 95% CIs. Code, processed datasets and execution instructions are provided in the appendix.~~Analyses used Python (pandas, statsmodels, linearmodels, PySAL, scikit-learn and LightGBM) (Rey & Anselin, 2007). Code, data and execution instructions are in the appendix and the study GitHub repository (https://github.com/VincentUwumboriyhieGmayinaam/NTD-Malaria-Burden-Financing). Tests were two-sided at α = .05.

Ethics and role of the funding source

~~The study used publicly available aggregate data without individual identifiers. The institutional determination regarding ethics review, study funding and the funder's role requires author confirmation.~~This study was based exclusively on secondary analyses of publicly available, aggregated data obtained from the Global Burden of Disease 2023 study, the World Bank's World Development Indicators, the World Health Organization through the United Nations Sustainable Development Goals database, the Institute for Health Metrics and Evaluation's Financing Global Health database, and Natural Earth. All data were available at the country or geographical level and contained no individual-level identifiable information. The study involved no direct recruitment of human participants, collection of primary data, or access to identifiable personal health records. Consequently, formal ethical approval and individual informed consent were not required for this secondary analysis, subject to applicable institutional requirements.

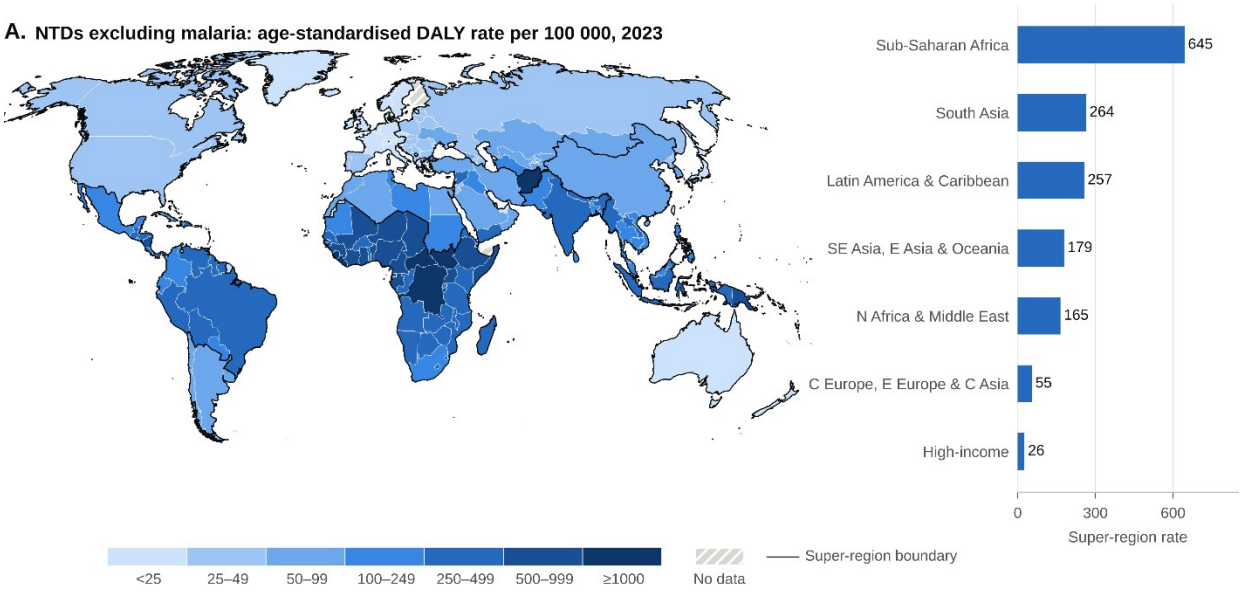
Results

~~Global and super-regional burden in 2023~~Global, regional and national burden

~~The supplied GBD global aggregate reports 70.83 million NTD and malaria DALYs in 2023 (95% UI 41.3–113.8 million). Separately, the 204 country estimates sum to 70.80 million, comprising 51.56 million malaria DALYs and 19.24 million NTD DALYs. Table 1 uses country-summed counts; the global UI comes from the supplied global aggregate and is not a sum of country bounds.~~In 2023, NTDs and malaria caused 70.83 million DALYs (95% UI 41.3–113.8 million); country estimates attributed 51.56 million to malaria and 19.24 million to NTDs (Table 1). Sub-Saharan Africa had 15.1% of the population but 80.2% of combined DALYs, 95.1% of malaria DALYs and 40.3% of NTD DALYs. Western sub-Saharan Africa alone accounted for 49.7% of global combined DALYs and central sub-Saharan Africa for 13.4%, with age-standardised rates of 5556.5 and 5006.5 per 100 000, against 314.2 in southern sub-Saharan Africa; malaria contributed 4828.5 and 4104.3 of those rates (Table S16; Figure S2). Outside Africa, Oceania had the highest rate (1694.8 per 100 000). For NTDs, South Asia held the largest regional share of DALYs (25.0%), followed by western (20.7%) and eastern sub-Saharan Africa (12.3%; Figure 2).

~~Sub-Saharan Africa accounted for 15.1% of the population but 80.2% of combined DALYs, including 95.1% of malaria DALYs and 40.3% of NTD DALYs. Its age-standardised NTD rate was 644.6 per 100 000, compared with 263.9 in South Asia, 256.8 in Latin America and the Caribbean, 179.4 in Southeast Asia, East Asia and Oceania, 165.2 in North Africa and the Middle East, 55.2 in Central Europe, Eastern Europe and Central Asia, and 26.5 in the high-income super-region. The malaria rate was 3237.1 per 100 000 in sub-Saharan Africa, 95.7 in South Asia and below 80 elsewhere.~~Three countries, India (4.00 million DALYs), Nigeria (2.10 million) and China (1.30 million), accounted for 38.5% of NTD DALYs, and Nigeria (16.63 million) and the Democratic Republic of the Congo (6.46 million) for 44.8% of malaria DALYs (Table S18). Rates were highest in South Sudan for NTDs (2227.1 per 100 000) and Sierra Leone for malaria (8237.2). Among the 19 NTDs, other NTDs (5.14 million DALYs), dengue (3.01 million) and schistosomiasis (2.07 million) contributed most (Figure 3; Table S2). Single countries held most of the burden of several causes: India 53.3% of lymphatic filariasis DALYs, China 75.6% of food-borne trematodiasis, Brazil 58.2% of Chagas disease and the Democratic Republic of the Congo 42.3% of onchocerciasis (Table S20).

~~The highest country-level NTD and malaria rates occurred across west, central and east Africa. Elevated NTD rates also occurred in South Asia, the Andean and Amazonian regions, and parts of Southeast Asia and the Pacific (Figure 2).~~Disability accounted for 62.4% of NTD DALYs (12.01 million YLDs), from 54.6% in sub-Saharan Africa to 94.5% in Central Europe, Eastern Europe and Central Asia, whereas premature death accounted for 95.2% of malaria DALYs (Table S17; Figure S3). GBD country rates correlated closely with WHO-reported people requiring NTD interventions (Spearman $\rho = 0.81$) and malaria incidence ($\rho = 0.94$; Table S21).



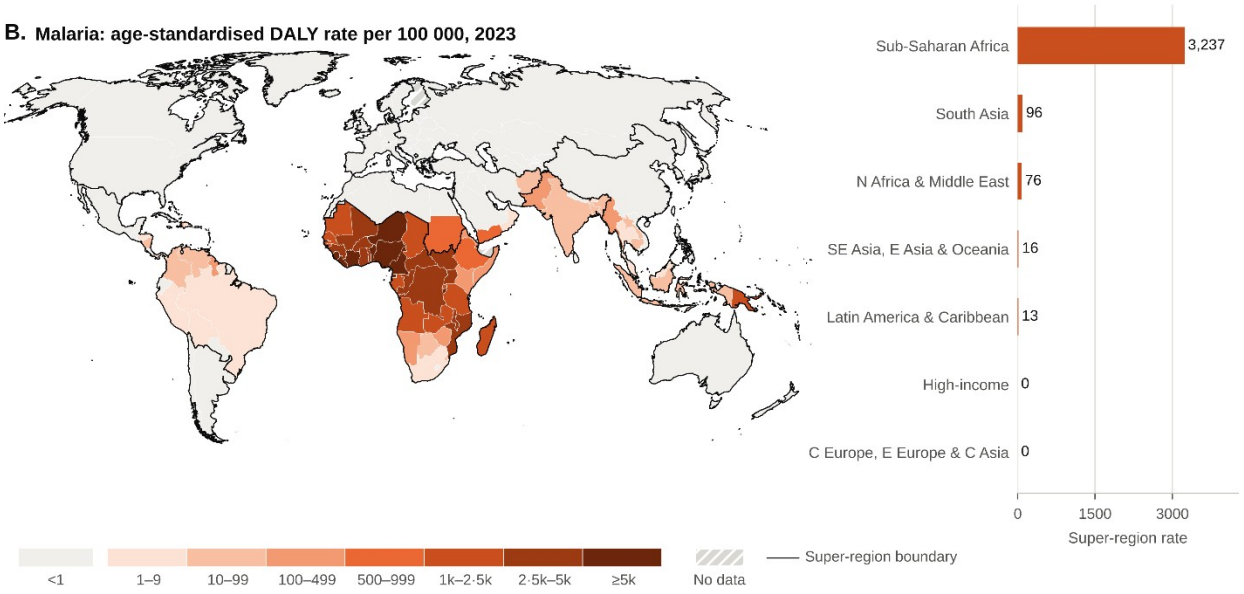


Figure 2: Geography of NTD and malaria burden, 2023. (A) Age-standardised DALY rate per 100 000 for NTDs excluding malaria and (B) malaria, by country, with GBD super-region boundaries; bars show super-region age-standardised rates. Stepped classes chosen to span the log-distributed rates. Hatched = no data.

~~The largest global NTD burdens were other NTDs (5.14 million DALYs), dengue (3.01 million), schistosomiasis (2.07 million), soil-transmitted helminthiases (1.54 million), lymphatic filariasis (1.49 million), cysticercosis (1.18 million), leishmaniasis (1.09 million) and food-borne trematodiases (1.04 million; Figure 3; Table S2). Sub-Saharan Africa accounted for all human African trypanosomiasis DALYs, 99% of onchocerciasis DALYs and 87% of schistosomiasis DALYs. Southeast Asia, East Asia and Oceania accounted for 86% of food-borne trematodiasis DALYs; Latin America and the Caribbean for 89% of Chagas disease DALYs; North Africa and the Middle East for 49% of leishmaniasis DALYs; and South Asia for 57% of lymphatic filariasis DALYs. South Asia and Southeast Asia together accounted for 81% of dengue DALYs.~~

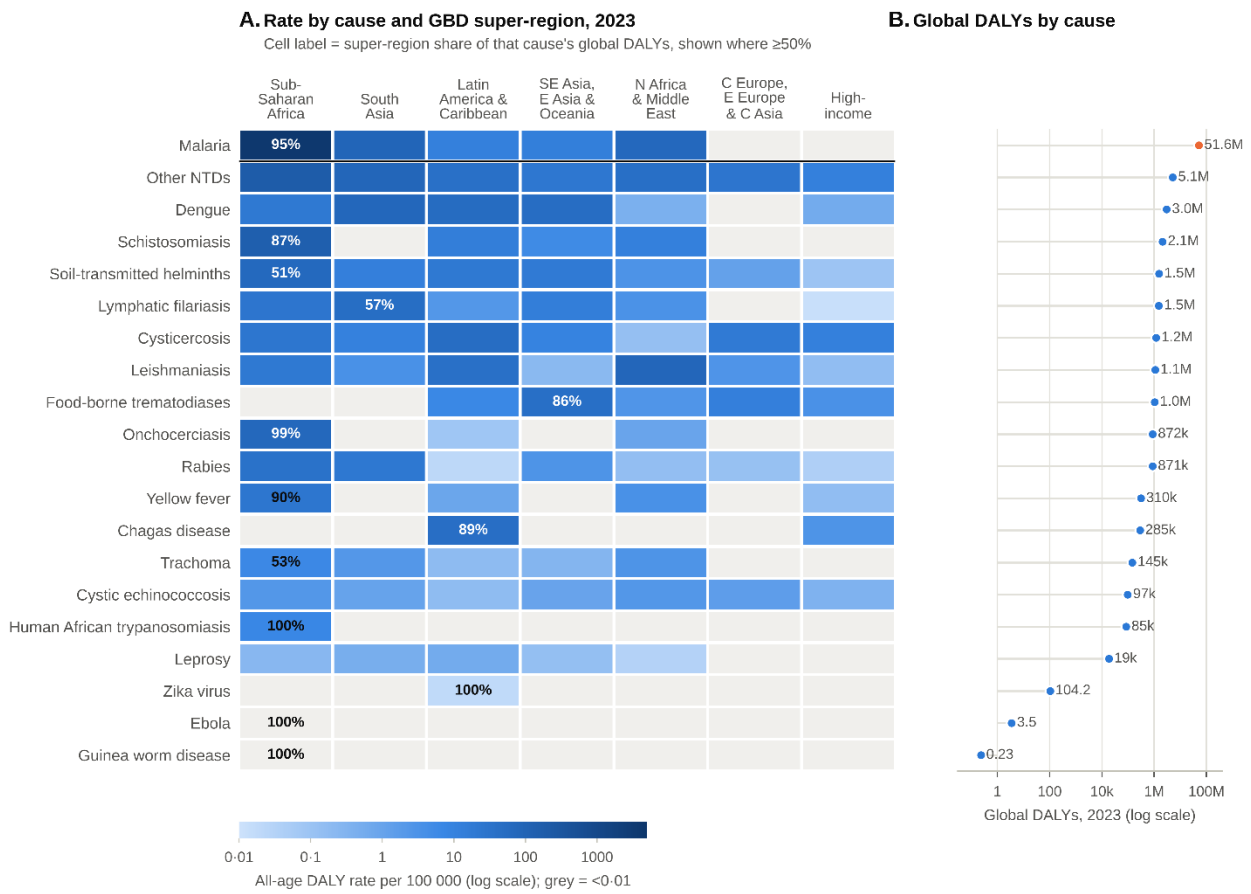


Figure 3: Cause-specific burden by GBD super-region, 2023. (A) All-age DALY rate per 100 000 by cause and super-region (log colour scale); labels show the super-region share of global DALYs where ≥50%. (B) Global DALYs by cause (log scale).

Temporal trends, 1990–2023

~~The global age-standardised NTD and malaria DALY rate declined from 1714.0 per 100 000 in 1990 to 1290.8 in 2010 and 1012.5 in 2023. The 2010–2023 reduction was 21.6% (95% UI 7.8–36.3; Table 1; Figure 4). The largest reductions occurred in South Asia (43.8%, 29.2–52.9), Southeast Asia, East Asia and Oceania (38.2%, 7.1–56.3), and sub-Saharan Africa (33.8%, 21.1–49.9). Point estimates declined in every super-region, although UIs included no change in Central Europe, Eastern Europe and Central Asia and the high-income super-region. In sub-Saharan Africa, the combined rate fell from 9711.4 in 1990 to 3881.7 in 2023, and its ratio to the global rate fell from 5.7 to 3.8. The malaria rate declined from 6929.5 to 3237.1. Globally, malaria rates initially rose from 1079.2 in 1990 to 1134.4 in 2005, then fell to 973.9 in 2010 and 765.7 in 2023. In South Asia, the combined rate fell from 1325.8 in 1990 to 359.6 in 2023; malaria rates fell from 582.8 to 95.7.~~The global age-standardised rate fell from 1714.0 per 100 000 in 1990 to 1290.8 in 2010 and 1012.5 in 2023, a fall of 21.6% (95% UI 7.8–36.3) since 2010 (Table 1; Figure 4). Since 2010, rates fell most in South Asia (43.8%, 29.2–52.9) and Southeast Asia, East Asia and Oceania (38.2%, 7.1–56.3), and by 34.1–39.9% in each of the four sub-Saharan African regions; sub-Saharan Africa's rate fell from 5.7 to 3.8 times the global rate between 1990 and 2023. Malaria rates rose only in North Africa and the Middle East (26.3%, 0.1–46.4) and fell to zero in Central Asia, Southern Latin America and Western Europe (Table S16). Nationally, 78 of 204 countries had a significant decline in the combined rate and four a significant increase: Nicaragua (105.0%, 46.2–178.8), Fiji (92.5%, 62.7–

132.2), Monaco (88.8%, 16.9–242.3) and American Samoa (56.2%, 20.4–91.4). Malaria rates were higher in 2023 than in 2010 in 13 countries, including Burundi, Madagascar, Zambia and Kenya (Table S19).

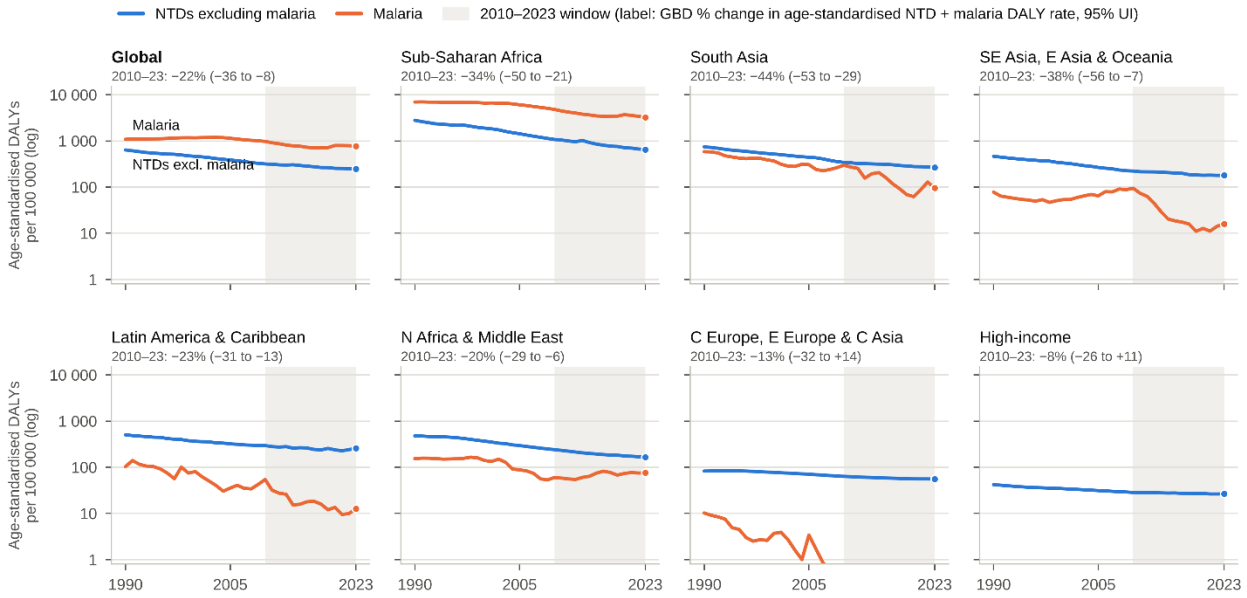


Figure 4: Trends in age-standardised DALY rates, 1990–2023. Small multiples by super-region on a shared log scale; shaded band marks 2010–2023, with GBD-published percentage change (95% UI) in the combined NTD and malaria rate.

Spatial clustering and socioeconomic inequality

~~Global Moran's I was 0.71 for NTDs (z = 19.3) and 0.79 for malaria (z = 21.1; both permutation p < .001; Table 2; Figure 5). Across alternative spatial weights, I ranged from 0.66 to 0.74 for NTDs and 0.73 to 0.82 for malaria. Clustering persisted within the sample outside the GBD high-income super-region (I = 0.67 and 0.77, respectively) and within malaria-endemic locations (I = 0.56; all p < .001; Table S7).~~Burden clustered geographically: Moran's I was 0.71 for NTDs and 0.79 for malaria (both p < .001; Table 2), and 0.66–0.82 across alternative weights (Table S7). High–high clusters (48 countries for NTDs and 42 for malaria) lay almost entirely in sub-Saharan Africa (Figure 5) and were stable: all 47 NTD high–high countries of 2010 remained so in 2023, with Somalia added, and Comoros and Somalia joined the malaria cluster (Table S21; Figure S4).

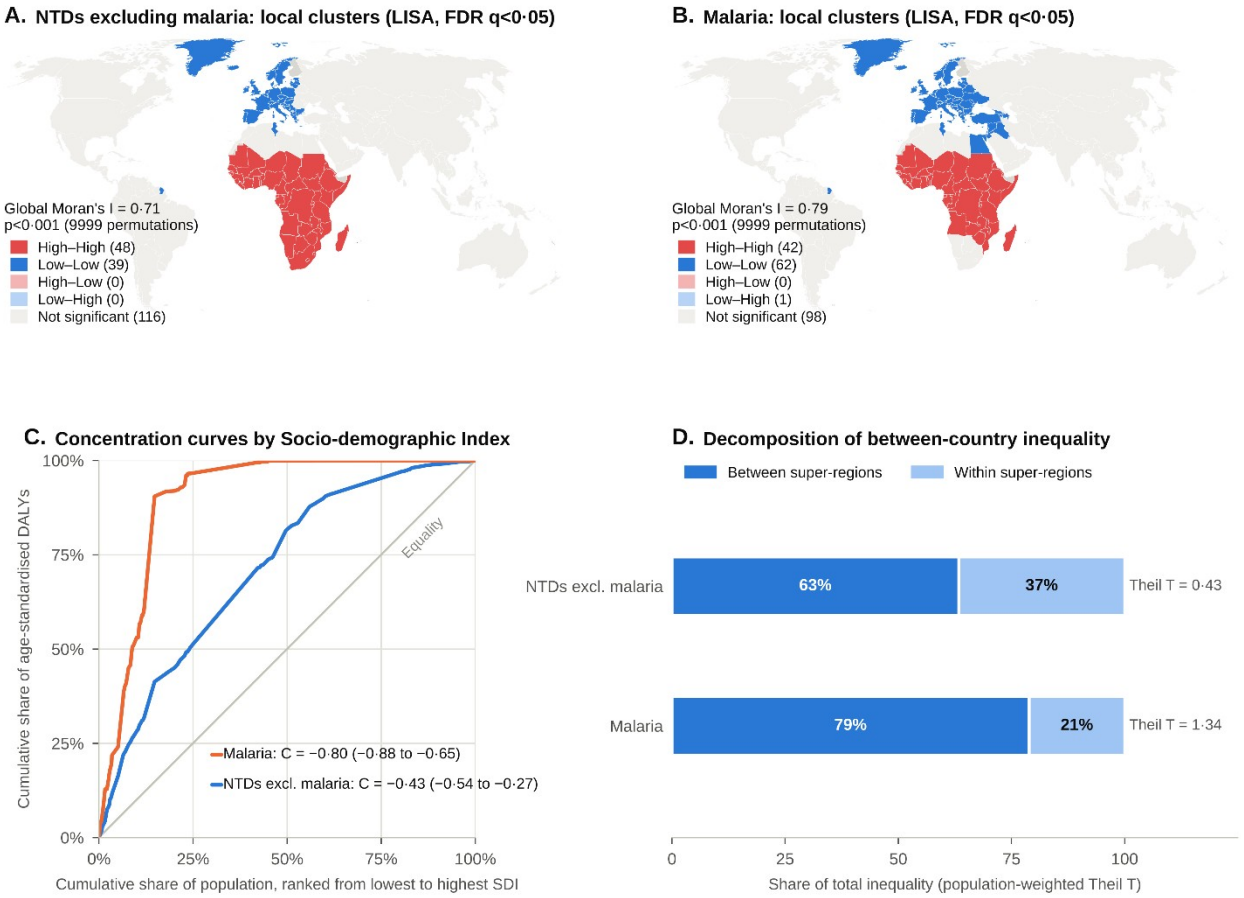


Figure 5: Spatial clustering and socioeconomic inequality. (A, B) LISA clusters (FDR q < .05) with global Moran's I; (C) concentration curves ranked by SDI with concentration indices (95% CI); (D) Theil T decomposition into between- and within-super-region inequality.

~~After false-discovery-rate adjustment, NTD burden showed 48 high–high clusters, 39 low–low clusters and no spatial outliers. Malaria showed 42 high–high clusters, 62 low–low clusters and one low–high outlier, Mauritius (Table 2). High–high clusters predominated in sub-Saharan Africa; 38 of 39 NTD low–low clusters were in the high-income super-region or Central Europe, Eastern Europe and Central Asia. Burden was concentrated in lower-SDI countries: concentration indices were −0.43 (95% CI −0.54 to −0.27) for NTDs and −0.80 (−0.88 to −0.65) for malaria. Theil T values were 0.43 and 1.34, respectively. Between-super-region differences accounted for 63.4% of NTD inequality and 78.9% of malaria inequality; within-super-region differences accounted for 36.6% and 21.1%. Including zero-rate locations increased malaria Theil T to 1.88, with 83.7% between-super-region inequality; the NTD result was unchanged (Table S15).~~Burden was concentrated in low-SDI countries, and relative concentration changed little over time. The malaria concentration index was −0.80 in 1990 (95% CI −0.88 to −0.63) and in 2023 (−0.88 to −0.65); the NTD index moved from −0.51 (−0.64 to −0.36) to −0.43 (−0.54 to −0.27), a change distinguishable from zero only for 2010–2023 (0.05, 0.02 to 0.08). Absolute inequality narrowed: the SII fell from −1966 (−3053 to −1198) to −632 (−877 to −412) DALYs per 100 000 for NTDs and from −3806 (−6754 to −1974) to −2484 (−4271 to −1131) for malaria (Table S15a; Figure S1). Most inequality lay between super-regions (79.3% for malaria and 64.0% for NTDs in 2023), but the share of NTD inequality between countries within the same region rose from 12.6% in 1990 to 21.1% in 2023 (Table S15b).

~~Burden and dependence on external financing~~Financing dependence and burden

~~Among 153 locations with financing data, median external expenditure share was 23.0% in sub-Saharan Africa, 11.5% in Southeast Asia, East Asia and Oceania, and 6.4% in South Asia. Medians were below 1% in the other three super-regions outside the high-income group. NTD burden correlated positively with external expenditure share across countries (Spearman ρ = 0.62; Figure 6). Tertile boundaries were 107 and 317 DALYs per 100 000 for burden and 0.9% and 12.4% for external share. Thirty-four countries were in the highest tertile of both measures: 31 in sub-Saharan Africa, two in North Asfrica and the Middle East, and one in Southeast Asia, East Asia and Oceania. They accounted for 27.2% of global NTD DALYs and 52.3% of malaria DALYs. Seventeen high-burden countries had low or intermediate external shares, including Brazil, Peru and Indonesia in the lowest financing tertile.~~Median external expenditure share was 23.0% in sub-Saharan Africa, 11.5% in Southeast Asia, East Asia and Oceania and 6.4% in South Asia. Thirty-four countries, 31 in sub-Saharan Africa, were in the highest tertiles of both NTD burden and external share; they held 27.2% of NTD DALYs and 52.3% of malaria DALYs (Figure 6). Seventeen high-burden countries, including Brazil, Peru and Indonesia, had low or intermediate external shares.

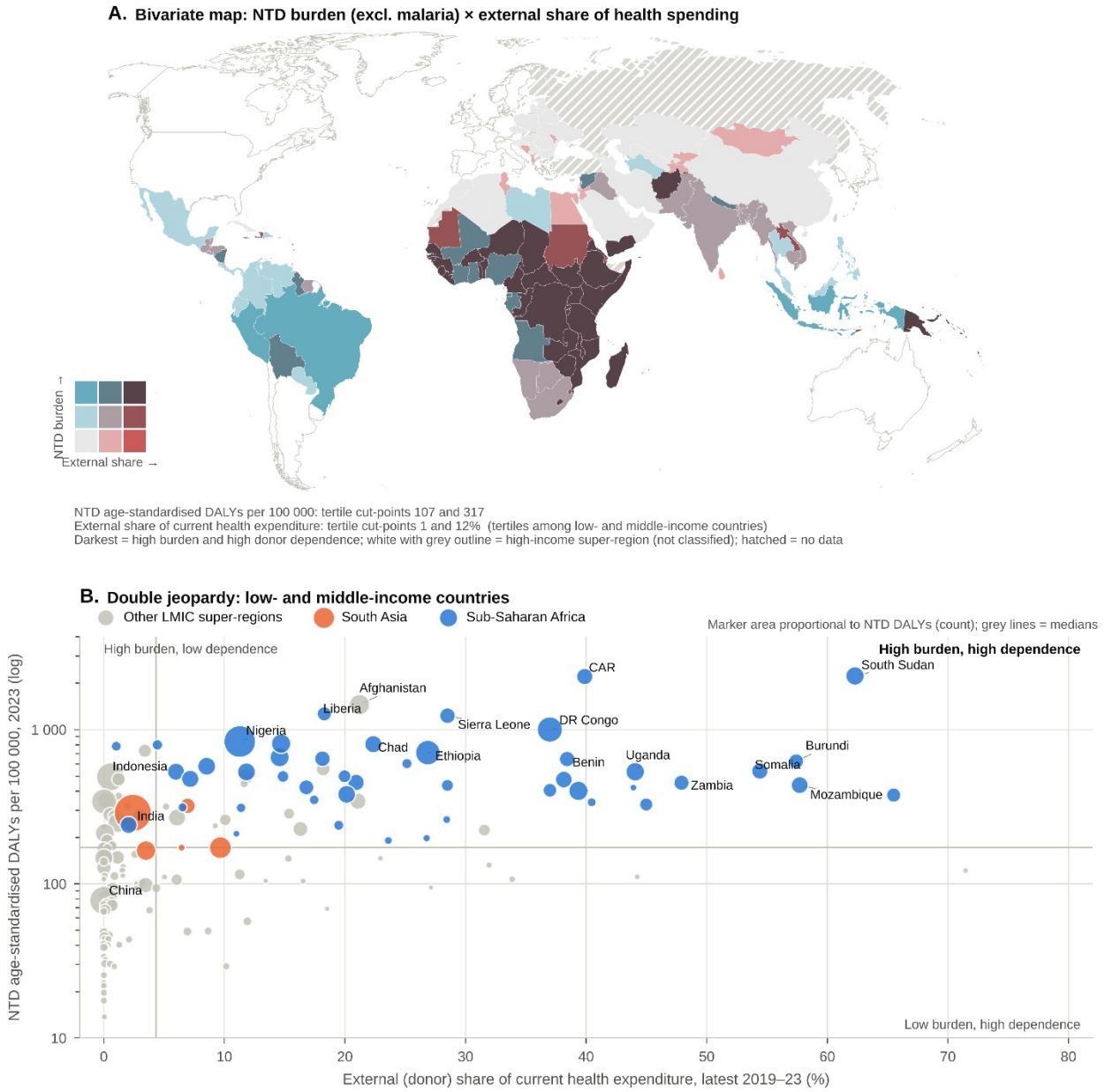


Figure 6: NTD burden and external health financing. (A) Bivariate tertile map of age-standardised NTD DALY rates and external expenditure share in the sample outside the GBD high-income super-region. (B) Country-level association, with marker area proportional to NTD DALYs and reference lines at the medians.

~~Associations between financing and burden~~

~~In the 2023 cross-section of 145 locations, a 10-percentage-point higher external expenditure share was associated with a 13.4% lower NTD DALY rate (95% CI 6.4–20.0), after adjustment for income, UHC, sanitation, drinking water, rural population share and super-region. The spatial error estimate was similar (−11.2%, −18.3 to −3.4; Table 3). The cross-sectional NTD association persisted in reported sensitivity analyses, including use of 2010–2014 financing averages and substitution of SDI for UHC and income (−16.3% per SD, 95% CI −26.2 to −5.1; Table S7). For the WHO-reported proportion requiring NTD interventions, the estimate was 8.4% per SD (−12.6 to 34.5; p = .46). Malaria cross-sectional estimates were imprecise and their CIs included zero. These sensitivity models did not all use the same eligible~~

~~population: most used global complete cases. They therefore do not isolate specification changes within the 145-location primary cross-section. For NTDs, diagnostics indicated country heterogeneity (Breusch–Pagan $\chi^2(1) = 15\,695$), rejection of random-effects assumptions (Hausman $\chi^2(6) = 59.7$; Mundlak $\chi^2(6) = 35.4$), cross-sectional dependence (Pesaran CD = 6.53), serial correlation (Wooldridge $F(1, 152) = 15.1$) and heteroskedasticity (modified Wald $\chi^2(153) = 647\,014$; all $p < .001$; Table S3). Malaria diagnostics were similar, although the Mundlak test did not reject random effects ($p = .12$). Fixed effects were retained for both outcomes, with Driscoll–Kraay standard errors in the preferred models.~~Between countries, a 10-percentage-point higher external share was associated with a 13.4% (95% CI 6.4–20.0) lower NTD rate in the 2023 cross-section (Table 3). Within countries, the association was −0.64% (−2.39 to 1.14) in the preferred model and ranged from −1.48% to 0.17% across within-country estimators, all intervals including zero (Figure 7A). In the correlated random-effects model, the between-country association was −4.99% (−11.25 to 1.71) and the within-country association −0.64% (Table S23). For malaria, the preferred within-country estimate was −6.32% in (1 + rate) (−10.85 to −1.56), but intervals included zero with country-clustered errors (−17.14 to 5.91) and in other within-country estimators, and the between-country association was positive (18.25%, −13.46 to 61.58). Malaria mortality gave a similar within-country estimate (−6.30%, −10.96 to −1.38; Table S24), as did alternative missing-data strategies and extension to 2023 (Tables S10b and S22). Higher UHC, sanitation and drinking-water coverage were each associated with lower NTD burden (Table S4). Local external-share coefficients from multiscale geographically weighted regression were significant in 36 of 145 countries, none in sub-Saharan Africa (Figure 7C; Table S6b).

~~In the preferred NTD model, a 10-percentage-point increase in external expenditure share was associated with a −0.64% difference in burden (95% CI −2.39 to 1.14; $p = .48$). Estimates across within-country models ranged from −1.48% for first differences to 0.17% for country-specific trends, with all CIs including zero. The per-capita specification gave −0.45% per doubling of expenditure plus US$1 (−1.08 to 0.19). The estimated association attenuated from −13.4% in the cross-section to −4.5% in pooled OLS and −0.6% in two-way fixed effects (Table 3; Figure 7A). For malaria, the preferred estimate was a −6.32% difference in rate + 1 per 10 percentage points (95% CI −10.85 to −1.56; $p = .010$). Estimates were sensitive to specification: clustered fixed effects, first differences, country trends and Mundlak models produced estimates from −1.95% to −6.41%, with all CIs including zero. The per-capita estimate was −4.4% per doubling of expenditure plus US$1 (−10.6 to 2.3). In the NTD model, UHC (−8.0% per 10 points, −9.1 to −7.0), sanitation (−4.6% per 10 percentage points, −5.7 to −3.4) and drinking-water coverage (−8.3%, −10.6 to −5.9) were associated with lower burden. Estimates for domestic government spending (−0.4%, −2.7 to 1.9) and income (5.1%, −0.7 to 11.2) included zero. In malaria models, domestic spending, income, sanitation and drinking water had negative coefficients; UHC had a positive coefficient (Table S4). With country-clustered inference, the same financing coefficient had a 95% CI of −17.14% to 5.91% ($p = .297$), showing that evidence against the null depended on the covariance estimator.~~

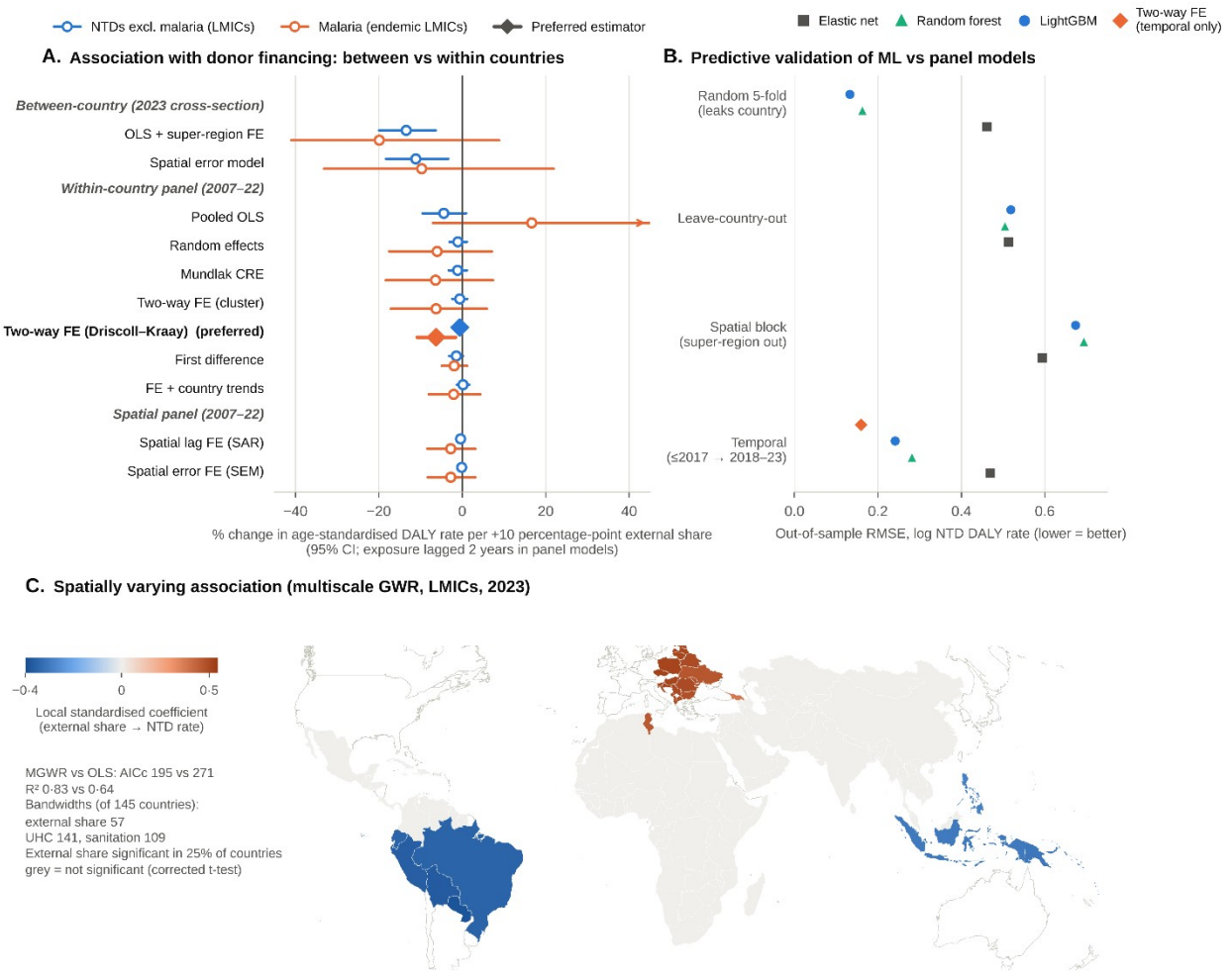


Figure 7: Donor financing and burden: estimator comparison, predictive validation and spatial heterogeneity. (A) Percentage change in age-standardised DALY rate per 10-percentage-point external share across between-country, within-country panel and spatial-panel estimators (preferred: two-way FE with Driscoll–Kraay SE). (B) Out-of-sample RMSE of machine-learning and panel models under four validation schemes. (C) Multiscale GWR local coefficients for external share (significant countries coloured).

~~Robustness to missing data handling~~

~~Excluding filled model inputs gave an NTD estimate of −0.69% (95% CI −2.46 to 1.11; 2225 observations) and a malaria estimate of −6.15% for rate + 1 (−10.83 to −1.21; 1173 observations). These estimates matched the strict observed-input analysis. Extending the panel through 2023 gave −0.32% (−2.13 to 1.53; 2429 observations) for NTDs and −6.97% (−11.28 to −2.45; 1262 observations) for malaria. Excluding the 186 NTD observations without matching raw UHC or LDI records gave −0.45% (−2.17 to 1.29); no malaria observations were affected. Omitting UHC gave −0.78% (−2.44 to 0.91) for NTDs and −6.59% (−11.62 to −1.28) for malaria. NTD confidence intervals included zero across bandwidth and country-clustered checks. Malaria intervals excluded zero with Driscoll–Kraay bandwidths of 1–4 but included zero with country-clustered inference (Tables S10b and S15).~~

~~Spatial panel models~~

~~MGWR fitted the 2023 NTD cross-section better than global OLS (AICc 194.9 vs 271.3; R^2 0.83 vs 0.64) and single-bandwidth GWR (AICc 219.6; Table S6a). Bandwidths were 141 countries for UHC, 109 for~~

~~sanitation, 66 for income and 57 for external share. UHC and sanitation coefficients were negative and significant throughout; income was significant in 33% of locations. External-share coefficients ranged from −0.39 to 0.53 (median −0.01) and remained significant after adjustment in 36 locations (25%): negative in 17, mainly in South America, Southeast Asia and the Pacific, and positive in 19, in central and eastern Europe and Tunisia. No sub-Saharan African location had a significant local coefficient (Table S6b; Figure 7C). The original predictive performance and SHAP summaries are retained in Table S5 and Figure 7B as provisional. The elastic-net internal tuning requires correction, and temporal comparisons require an identical test sample across models. Observed test-period covariates make these conditional predictions. A ranking of the models is therefore withheld pending the corrected validation.~~

Projected health losses under development-assistance cuts

~~The econometric model projected 25 000 excess NTD DALYs (−42 000 to 90 000) and 1.22 million excess malaria DALYs (0.27–2.29 million; Figure 8A–B). Under the accounting model, mild cuts projected 108 000 excess NTD DALYs (95% UI 55 000–171 000) and 825 000 malaria DALYs (434 000–1 244 000). Severe cuts projected 540 000 NTD DALYs (404 000–718 000) and 4.10 million malaria DALYs (3.26–5.07 million), equivalent to 2.9% and 8.0% of baseline burden, respectively.~~Under the reference scenario, the accounting model projected 406 600 excess NTD DALYs a year (95% UI 334 500–501 800) and 1.85 million excess malaria DALYs (1.51–2.21 million), 2.2% and 3.6% of 2023 burden; severe cuts projected 908 700 (708 000–1 140 400) and 4.10 million (3.25–5.07 million; Table 4; Figure 8A–B). The econometric model projected 24 500 excess NTD DALYs (−42 100 to 90 300) and 1.22 million malaria DALYs (0.27–2.29 million). Sub-Saharan Africa accounted for 91% of excess NTD and 96% of excess malaria DALYs in the reference accounting scenario. The largest NTD projections were for the Democratic Republic of the Congo (109 900; 83 600–145 100), Ethiopia (34 700) and Tanzania (32 000), and the largest per-capita projections for South Sudan (246 per 100 000) and the Democratic Republic of the Congo (104; Table S8; Figure 8C). Reference-scenario losses were valued at US$0.55 billion (0.45–0.72) for NTDs and US$2.11 billion (1.65–2.55) for malaria a year. Replacing lost external funds would require domestic government health expenditure to rise by 8.7% in sub-Saharan Africa (US$3.44 billion; 3.9–19.4% across scenarios), 1.8% in South Asia and 0.5% in North Africa and the Middle East (Table S14; Figure 8D). Independent burden draws widened intervals without materially changing medians (Table S7).

~~Sub-Saharan Africa accounted for 94% of projected excess NTD DALYs and 96% of excess malaria DALYs in the reference accounting scenario. The largest absolute NTD projections were for the Democratic Republic of the Congo (95 800; 95% UI 70 400–129 200), South Sudan (23 800), Uganda (21 300), Tanzania (20 800) and Ethiopia (15 000; Table S8). Projected NTD losses per 100 000 population were highest in South Sudan (207), the Democratic Republic of the Congo (91), Uganda (44), Tanzania (31), Liberia (30) and Mozambique (29; Figure 8C). External expenditure accounted for 18–62% of CHE in these countries. At one time GDP per capita per DALY, reference-scenario health losses were valued at US$0.32 billion annually for NTDs (95% UI 0.24–0.47) and US$2.11 billion for malaria (1.65–2.55). Values ranged from US$0.16–0.95 billion for NTDs and US$1.05–6.33 billion for malaria under the 0.5–3 times GDP-per-capita assumptions (Table S7). Replacing external financing lost under the reference scenario would require domestic government health expenditure to increase by 8.7% in sub-Saharan Africa (US$3.44 billion), 1.8% in South Asia (US$0.94 billion) and 0.5% in North Africa and the Middle East (US$0.64 billion). Increases were 0.11% or less in the other study super-regions. For sub-Saharan Africa, the required increase ranged from 3.9% under mild cuts to 19.4% under severe cuts (Table S14; Figure 8D).~~

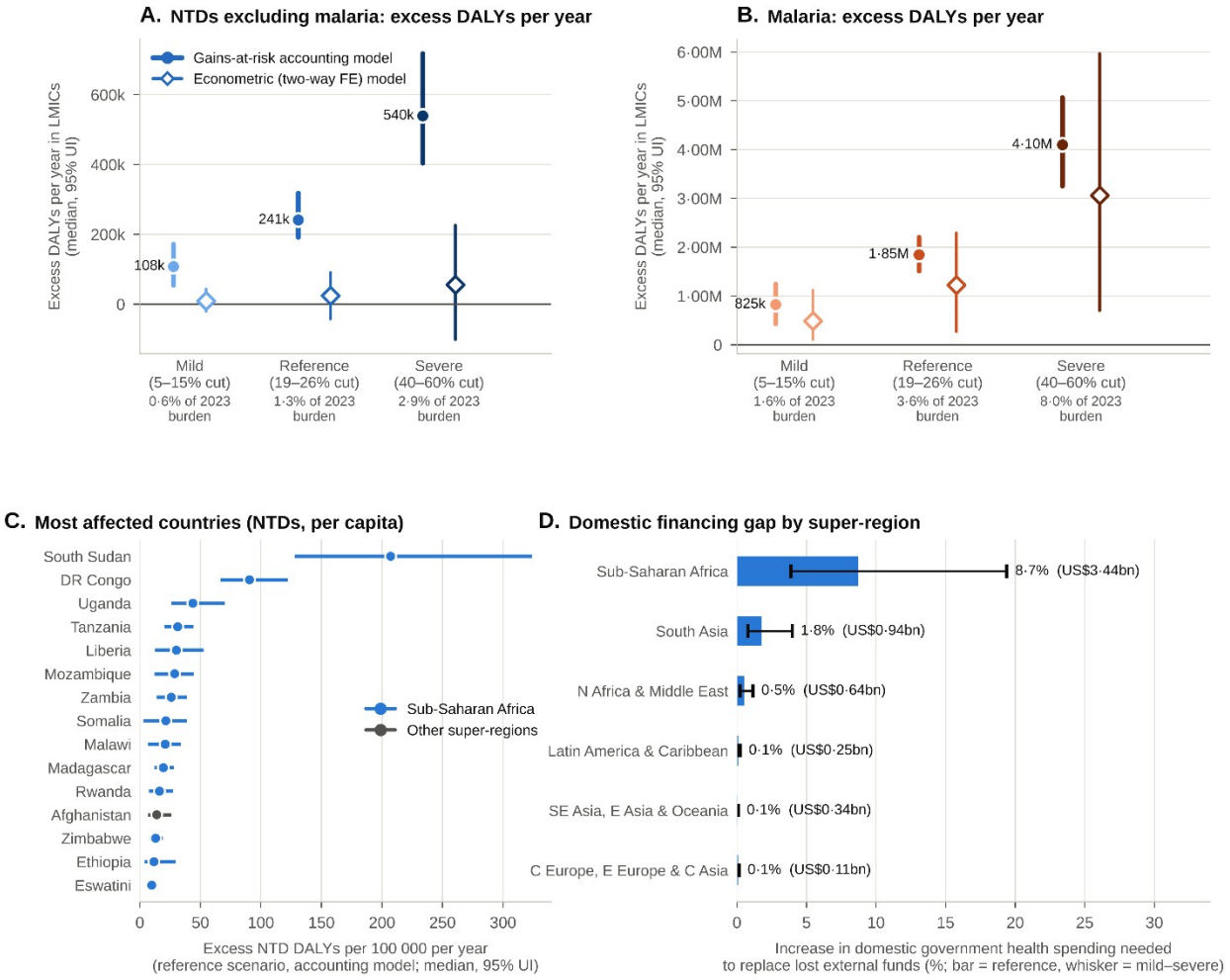


Figure 8: Projected consequences of development-assistance cuts. (A, B) Excess annual DALYs in eligible financing-scenario locations under three cut scenarios from the accounting (filled) and econometric (open) models, median and 95% UI; (C) countries with the largest per-capita excess NTD burden (reference scenario); (D) increase in domestic government health spending needed to replace lost external funds, by super-region

~~Scenario sensitivity analyses~~

~~The original independent-draw sensitivity and the reported 80% USAID-attribution result remain unverified. The attribution script reads an earlier scenario model and aggregates country-level interval bounds. It must be regenerated from a supported accounting dataset, with totals calculated within each simulation draw before interval estimation. The earlier estimate of 536 000 excess NTD DALYs is therefore not retained as a substantive result.~~

Discussion

~~Sub-Saharan Africa carried 80.2% of the combined NTD and malaria DALY burden in 2023, despite a 33.8% reduction in its age-standardised burden rate since 2010. Burden was concentrated in lower-SDI countries, and 34 countries with both high NTD burden and high external financing dependence accounted for over half of global malaria DALYs. The financing analyses, however, yielded a more qualified result: a substantial adjusted between-country NTD association attenuated towards zero when estimated from changes within countries, while the malaria association depended on the specification and~~

~~covariance estimator. Together, these findings identify where epidemiological need and financing vulnerability coincide, while showing why this overlap cannot establish the magnitude of health losses attributable to financing withdrawal.~~NTDs and malaria caused 70.8 million DALYs in 2023, 63.1% of them in western and central sub-Saharan Africa. Since 1990 the absolute socioeconomic gap narrowed by about two-thirds for NTDs and one-third for malaria, but burden remained as concentrated in low-SDI countries as before and hot spots did not move. Thirty-four countries combining high NTD burden with high donor dependence held over half of malaria DALYs. The inverse association between external financing and NTD burden seen across countries was not detectable within countries over time, and the malaria association depended on the estimator. These findings locate where need and financing vulnerability coincide; they do not establish the size of losses attributable to withdrawal.

~~The continuing concentration of burden in sub-Saharan Africa is consistent with recent GBD 2021 analyses, which reported persistent disadvantage in lower-SDI settings despite declining global age-standardised rates (He et al., 2025; Lin et al., 2025). Our analysis extends the observation period to 2023 and distinguishes the geography of malaria from that of the NTD aggregate: sub-Saharan Africa accounted for 95.1% of malaria DALYs but 40.3% of NTD DALYs. Agreement across these analyses should be interpreted as consistency within successive GBD estimation frameworks, rather than independent replication using unrelated surveillance data (GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025; He et al., 2025; Lin et al., 2025). The distinction between disease groups matters for allocation: our cause-specific results place substantial lymphatic filariasis burden in South Asia and Chagas disease burden in Latin America and the Caribbean, which a combined global burden ranking could obscure. These distributions support disease-specific priorities within a broader commitment to geographical equity (WHO, 2020).~~Our 2023 total is close to the 71.6 million DALYs reported for 2021 from GBD 2021 (Zheng et al., 2025); because each GBD round re-estimates the full series, the difference does not measure change. The regional pattern agrees with GBD 2019 and 2021 analyses identifying western sub-Saharan Africa as the highest-burden region (Feng et al., 2025; Lin et al., 2025), and stable high–high clusters extend malaria-only geospatial evidence (Namubiru, 2025) to NTDs. National concentration matters for planning: three countries held 38.5% of NTD DALYs and two held 44.8% of malaria DALYs, and single countries held most of the burden of lymphatic filariasis, food-borne trematodiasis and Chagas disease, which aggregate rankings conceal. The post-2015 plateau in malaria incidence in Africa (Weiss et al., 2025) and higher malaria rates in 13 countries in 2023 than in 2010 show that recent progress has been neither uniform nor secure.

~~The 21.6% global reduction in the combined age-standardised DALY rate during 2010–2023 does not imply uninterrupted progress throughout that period. In their 2000–2022 geospatial analysis, Weiss et al. (2025) found that malaria infection prevalence and incidence in sub-Saharan Africa had plateaued after 2015, although mortality continued to improve outside the pandemic-affected years. This apparent difference reflects the comparison of a long-period change in combined DALYs with more recent trends in malaria-specific incidence and mortality; the estimates address different outcomes and time intervals (Weiss et al., 2025). Earlier intervention modelling attributed 663 million averted clinical malaria cases during 2000–2015 to control measures, with insecticide-treated nets contributing the largest share, demonstrating the scale of gains that preceded the more recent plateau (Bhatt et al., 2015). Our finding that sub-Saharan Africa's burden rate remained 3.8 times the global rate therefore indicates substantial residual inequality alongside historical improvement. Tracking both rates and absolute burden is necessary because population growth can increase service requirements even when individual risk declines (Weiss et al., 2025).~~

~~Spatial clustering and socioeconomic concentration described related, but distinct, dimensions of inequality. Moran's I values of 0.71 for NTDs and 0.79 for malaria indicated geographical clustering, whereas concentration indices of −0.43 and −0.80 located burden disproportionately among lower-SDI populations. The direction agrees with Tusting et al.'s (2013) systematic review, in which children in the poorest households had approximately twice the adjusted odds of malaria infection compared with those in the least-poor households. That household-level association cannot be compared numerically with national DALY concentration indices, and the review's observational evidence did not establish the direction of causation (Tusting et al., 2013). Our decomposition adds the distinction between inequalities within and between super-regions, although the between-region malaria contribution increased from 78.9% to 83.7% when zero-rate locations were retained. This sensitivity shows that the magnitude depends partly on the population included in the comparison. Moreover, country-level clustering does not demonstrate cross-border transmission or identify district-level intervention sites; finer-resolution malaria mapping illustrates the additional information needed for local targeting (Weiss et al., 2025).~~Absolute and relative inequality moved differently. Narrowing absolute gaps reflect falling rates in low-SDI countries, but unchanged concentration indices show that these countries still carry the same disproportionate share of burden; GBD 2021 analyses found relative inequality widening to 2021 (Feng et al., 2025; Lin et al., 2025), and our estimates to 2023 show at most stabilisation. Targets framed in absolute terms can therefore be met while the relative disadvantage of the poorest countries persists (Mackenbach & Kunst, 1997). The rising within-region share of NTD inequality suggests that residual NTD burden increasingly reflects differences between neighbouring countries, consistent with uneven subnational treatment coverage (Khaki et al., 2025; Lo et al., 2019), whereas malaria inequality remains largely between super-regions. Associations of NTD burden with sanitation and water coverage are consistent with evidence on water, housing and cash transfers (Ahmed et al., 2022; Grimes et al., 2014; Tusting et al., 2017), but do not identify intervention effects.

~~The inverse associations of NTD burden with sanitation and drinking-water coverage provide a plausible connection between socioeconomic disadvantage and transmission, without identifying the effects of specific infrastructure interventions. A systematic review associated access to safe water with lower odds of schistosomiasis, but its predominantly cross-sectional, generally poor-quality evidence limits causal interpretation (Grimes et al., 2014). Similarly, a 21-country analysis associated improved housing with lower childhood malaria infection after adjustment for wealth and malaria interventions, supporting a role for living conditions beyond national income alone (Tusting et al., 2017). Evidence also extends beyond infrastructure: Ahmed et al. (2022) identified improved NTD outcomes in seven of 11 studies of conditional cash transfers, particularly for leprosy and deworming uptake. Those programmes differed in eligibility, conditions and outcomes, so their results do not establish a common effect across the 19 NTD causes analysed here (Ahmed et al., 2022). Our findings consequently support investigating coordinated disease-control and social-protection approaches, while leaving their comparative effectiveness to intervention-specific evaluations.~~

~~The 34 countries in the highest tertiles of NTD burden and external financing share accounted for 27.2% of global NTD DALYs and 52.3% of malaria DALYs, identifying a concentration of potential exposure to financing disruption. This overlap is consistent with the large disparities in health resources documented by the Global Burden of Disease 2021 Health Financing Collaborator Network (2023), but our classification adds the joint distribution of disease burden and financing dependence. A high external share can reflect substantial donor support, limited domestic expenditure, or both, because the indicator is a ratio rather than a measure of programme resources (World Bank, n.d.). Nor does the positive unadjusted correlation between burden and financing share establish that assistance increases burden: allocation towards countries with greater need is a plausible explanation that our descriptive analysis~~

~~cannot distinguish from other country differences. The tertile classification is therefore a screening tool for closer programme and fiscal assessment, not a causal vulnerability score or a threshold for withdrawing support. The 17 high-burden countries with lower external shares also show why donor dependence alone would miss important disease-control needs.~~

~~The NTD financing association changed from a 13.4% lower DALY rate per 10-percentage-point higher external share in the adjusted cross-section to a −0.64% within-country difference (95% CI −2.39 to 1.14). This attenuation is informative because cross-sectional and fixed-effects models estimate different relationships: the former compares countries, whereas the latter uses temporal variation after removing stable country characteristics and common year effects (Bell & Jones, 2015; Imai & Kim, 2021). Earlier multicountry studies associated health aid with improved infant mortality, under-five mortality or life expectancy, using absolute or per-capita assistance measures and different periods (Bendavid & Bhattacharya, 2014; Mishra & Newhouse, 2009). More recently, Cavalcanti et al. (2025) reported lower malaria and NTD mortality with higher USAID support in 133 countries and territories during 2001–2021, while A. F. da Silva et al. (2026) reported similar directions for official development assistance in 93 countries during 2002–2021. These findings contrast with our statistically inconclusive NTD result, but mortality, combined fatal and non-fatal DALYs, donor-specific support, broad development assistance and external health-expenditure shares are not interchangeable outcomes or exposures (Cavalcanti et al., 2025; A. F. da Silva et al., 2026). The recent analyses also share investigators and related modelling approaches, so their agreement should not be treated as wholly independent methodological confirmation.~~The NTD financing association attenuated from 13.4% between countries to 0.64% within countries, and the Mundlak model reproduced this contrast in one sample. Earlier panels associated aid volume with lower malaria and child mortality (Akachi & Atun, 2011; Bendavid & Bhattacharya, 2014; Hsiao & Emdin, 2014; Yan et al., 2015), and recent analyses associated USAID and total aid with lower NTD and malaria mortality (Cavalcanti et al., 2025; A. F. da Silva et al., 2026). Our exposure measures dependence rather than volume, and our outcome includes disability, which makes up 62.4% of NTD DALYs and which mortality analyses cannot capture. For malaria, within-country associations with DALYs and deaths were negative, in the direction of programme evaluations (Bhatt et al., 2015; Jakubowski et al., 2017), whereas the between-country association was positive, consistent with aid allocation to high-burden countries (Namubiru, 2025; Sempungu et al., 2023). This contrast is why between-country coefficients should not be converted into projected losses. The weak NTD association may reflect an exposure that does not isolate NTD funding, which received 0.6% of health aid in 2003–2007 (Liese & Schubert, 2009), aggregation of diseases with different response times (Borlase et al., 2022), and a fixed 2-year lag; these are hypotheses that this study cannot test.

~~Several features could contribute to the weak within-country NTD association. Our exposure combines externally financed health activities and does not isolate expenditure on NTD treatment, surveillance or morbidity management; a change in its denominator can alter the share without changing programme funding (World Bank, n.d.). Aggregating heterogeneous NTDs may also conceal disease-specific responses, while transmission models show that the consequences of interrupted control depend on the disease, endemicity and programme history (Borlase et al., 2022). The assumed two-year lag may not represent these different response times, and lagging an endogenous exposure does not itself remove reverse causation or time-varying confounding (Bellemare et al., 2017). Adjustment for health-service coverage could additionally remove part of a financing pathway if coverage mediates the relationship, although omitting UHC in our sensitivity analysis left the NTD estimate close to zero. These explanations remain hypotheses rather than mechanisms demonstrated by this study. The contribution is the explicit separation of between-country and within-country evidence, which cautions against translating the stronger cross-sectional coefficient directly into projected health losses.~~

~~For malaria, a 10-percentage-point increase in external share was associated with a 6.32% reduction in rate plus one under the preferred model, but the confidence interval included zero with country-clustered inference. The outcome transformation means that this coefficient is not exactly a 6.32% reduction in the untransformed DALY rate, particularly where rates are low. The negative direction is compatible with Jakubowski et al. (2017), who associated PMI introduction with a 16% reduction in annual under-five mortality risk and increased coverage of malaria prevention in a difference-in-differences comparison of 19 recipient and 13 non-recipient countries. Their programme-specific exposure and household-derived mortality outcome are more closely linked to an identifiable intervention than our aggregate financing ratio, although their observational design also allowed residual confounding (Jakubowski et al., 2017). Bhatt et al. (2015) further support the importance of intervention coverage, but their counterfactual attribution model does not identify a marginal effect of external expenditure. Our result therefore provides suggestive, specification-sensitive evidence and strengthens the case for linking expenditure to actual service coverage before estimating the consequences of reductions.~~

~~The scenario analyses address a different question from the historical regressions: how burden might change under explicitly specified financing reductions. Their numerical health-loss estimates remain provisional because the accounting analysis lacks five historical NTD cause baselines and the malaria econometric calculation requires a corrected conversion from log-transformed rates. Consequently, the retained projections cannot yet establish the magnitude of excess DALYs. The accounting approach assumes that the externally financed share of expenditure represents the same share of historical health gains and that those gains reverse proportionately; the econometric approach instead transfers an estimated historical association to a hypothetical financing change. Neither assumption follows from the observed distribution of burden. Presenting the approaches separately is useful because it makes structural uncertainty visible, but narrower simulation intervals cannot compensate for an unsupported baseline or an incorrect outcome conversion (Briggs et al., 2012). The reference cut range is informed by Apeagyei et al.'s (2025) projected contraction in global development assistance for health, rather than evidence that every country or NTD programme would experience an identical reduction.~~The scenario models bracket structural uncertainty. The accounting model assumes that donor-financed spending produced a matching share of health gains; the econometric model transfers a weak historical association; their NTD projections differ 17-fold. Neither represents the nonlinear rebound that follows interrupted mass drug administration (Borlase et al., 2022) or the resurgence that followed weakened control in 68 of 75 malaria resurgence events (Cohen et al., 2012), and the programme-specific malaria projections of Symons et al. (2025) address a different counterfactual. Replacing lost funds would require an 8.7% rise in domestic government health spending in sub-Saharan Africa. Reductions in health aid have not historically been offset by domestic spending (Dieleman & Hanlon, 2014), aid has had no significant effect on overall domestic public health spending (Patenaude, 2021), poorly prepared donor transitions disrupt services (Huffstetler et al., 2022), and higher tax revenue is the main route to greater public health spending (Reeves et al., 2015).

~~Published disruption models provide relevant comparators without validating these provisional magnitudes. Symons et al. (2025) estimated that planned PMI activities could avert 13.6 million malaria cases and 104,000 deaths in 2025 relative to withdrawal without mitigation; this programme-specific counterfactual differs from our proportional reductions in total external health expenditure. Borlase et al. (2022) modelled pandemic-related interruptions across seven NTDs and found that a one-year interruption could delay elimination targets by two to three years for several diseases in high-prevalence settings. Their results indicate that interruption effects can be nonlinear and dependent on timing, which a proportional accounting model cannot represent (Borlase et al., 2022). Historical evidence is consistent with this concern: programme weakening was implicated in 68 of 75 malaria resurgence events reviewed~~

~~by Cohen et al. (2012), although retrospective attribution cannot isolate a causal financing effect. The WHO regional assessment also documented disruption of NTD activities following funding reductions, but reported implementation experience and anticipated cancellations rather than measured excess DALYs (World Health Organization Regional Office for Africa, 2025). Taken together, these sources support service-continuity planning while underscoring the need to distinguish observed interruption, modelled health consequences and uncertainty about their scale.~~

~~Replacing the reference-scenario financing loss would require an estimated 8.7% increase in domestic government health expenditure in sub-Saharan Africa, compared with 1.8% in South Asia and 0.5% in North Africa and the Middle East. These are conditional budget requirements, not estimates of resources that governments can mobilise. Historical evidence found no statistically detectable domestic compensation following reductions in health aid, while a separate 89-country analysis associated greater tax revenue with higher government health expenditure (Dieleman & Hanlon, 2014; Reeves et al., 2015). Neither study establishes the feasibility of replacing current losses, but both indicate why expenditure needs must be assessed alongside revenue generation and budget allocation (Dieleman & Hanlon, 2014; Reeves et al., 2015). Evidence from 89 publications on donor transitions identified financial, workforce, procurement and service-delivery problems when preparation was inadequate; much of that literature concerned HIV and immunisation in middle-income countries, limiting direct transfer to NTD programmes in low-income settings (Huffstetler et al., 2022). Comparative evidence from China, Georgia, Sri Lanka and Uganda similarly emphasised integration into domestic financing and service-delivery systems, rather than financial substitution alone (Shroff et al., 2024). Our regional estimates provide a starting point for country-specific transition costing and negotiations over the timing of reductions.~~

~~The study brings together burden distributions, socioeconomic inequality, financing dependence and alternative longitudinal specifications in a common analytical framework. Its strongest contribution is the finding that substantial geographical and financial vulnerability can coexist with weak or specification-sensitive estimates of the financing–burden relationship. Several limitations define the interpretation. National averages cannot establish individual-level effects or locate underserved communities within countries (Morgenstern, 1995). GBD outcomes are modelled estimates whose precision and comparability depend on the underlying evidence and estimation procedures; analyses using related GBD releases are therefore not independent validation datasets (GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025). The financing samples were smaller than the descriptive sample, and complete-case exclusions, interpolation and the absence of full imputation diagnostics restrict the interpretation of sensitivity results (Sterne et al., 2009). Country and year fixed effects do not remove time-varying confounding, and our nearest-neighbour spatial weights describe geographical proximity rather than observed transmission connections (Anselin, 1988; Imai & Kim, 2021). The provisional spatial-panel, prediction and scenario outputs consequently cannot strengthen the substantive conclusions until their documented implementation problems are resolved.~~This ecological study cannot establish individual-level effects or locate underserved communities within countries (Morgenstern, 1995). GBD outcomes are modelled; close agreement with WHO series supports the ordering of countries, not absolute levels. Financing samples were smaller than the descriptive sample, missing WDI values were interpolated, and fixed effects do not remove time-varying confounding. Country-level changes for NTDs excluding malaria and for malaria alone lack uncertainty intervals, and the spatial-panel and machine-learning analyses remain provisional (appendix).

~~The evidence supports prioritising continuity of NTD and malaria services where high burden coincides with limited domestic financing, while adapting priorities to each disease's geographical distribution~~

~~(WHO, 2020; Huffstetler et al., 2022). Our findings identify countries and regions requiring closer assessment, but do not justify treating external-financing shares as disease-specific causal effects. The next analytical step is to link verified programme disbursements with coverage, interruption duration and disease-specific outcomes, informed by the intervention and transmission models used in existing research (Bhatt et al., 2015; Borlase et al., 2022; Symons et al., 2025). Protecting services during financing transitions is supported by the broader evidence; the precise excess burden attributable to future cuts requires corrected calculations and more direct measurement of the programmes affected.~~NTD and malaria burden remains concentrated where poverty and donor dependence are greatest, and the 2025 cuts fall on those countries. These data support protecting service continuity there, with priorities adapted to each disease's geography. Measuring the burden attributable to the cuts requires linking programme disbursements to coverage and disease-specific outcomes.

CONTRIBUTORS (Required)

Detail each author's specific contributions (which needs to match your signed Author statement forms); confirm all authors had full access to the data and final responsibility for submission; identify at least two authors who have accessed and verified the underlying data, including at least one from the academic team where applicable.

~~DECLARATION OF INTERESTS (Required)~~DECLARATION OF INTERESTS

~~*Declare any competing interests for all authors. The declaration should match your signed ICMJE forms – eg, "XX declares grants from XX. All other authors declare no competing interests."*~~

All other authors declare no competing interests.

~~DATA SHARING (Required)~~DATA SHARING

~~*Provide a data sharing statement describing whether individual participant data will be shared; what data; when; with whom; and how to access.*~~

All data analysed in this study were obtained from publicly accessible secondary data repositories. Disease burden, sociodemographic, and universal health coverage estimates were retrieved from the Global Burden of Disease (GBD) 2023 study through the Institute for Health Metrics and Evaluation (IHME) (https://vizhub.healthdata.org/gbd-results/). Economic and health-system indicators were obtained from the World Bank's World Development Indicators database (https://databank.worldbank.org/source/world-development-indicators). Sustainable Development Goal indicators were obtained from the United Nations SDG Global Database, incorporating data reported by the World Health Organization (https://unstats.un.org/sdgs/dataportal). Development assistance for health estimates were obtained from IHME's Financing Global Health database (https://www.healthdata.org/data-tools-practices/data-sources), and geographical boundary data were sourced from Natural Earth (https://www.naturalearthdata.com/). The analytical dataset was constructed by harmonising and linking these sources using country identifiers and calendar years. All original datasets are accessible through their respective repositories, subject to applicable data-use conditions. The harmonised analytical dataset and analytical code are held in the study GitHub repository (https://github.com/VincentUwumboriyhieGmayinaam/NTD-Malaria-Burden-Financing), which will be made publicly available on publication.

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References

Ahmed, A., Aune, D., Vineis, P., Pescarini, J. M., Millett, C., & Hone, T. (2022). The effect of conditional cash transfers on the control of neglected tropical disease: A systematic review. *The Lancet Global Health*, *10*(5), e640–e648. https://doi.org/10.1016/S2214-109X(22)00065-1

Akachi, Y., & Atun, R. (2011). Effect of investment in malaria control on child mortality in sub-Saharan Africa in 2002–2008. *PLoS ONE*, *6*(6), Article e21309. https://doi.org/10.1371/journal.pone.0021309

~~Anselin, L. (1988). *Spatial econometrics: Methods and models*. Kluwer Academic Publishers.~~

Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical Analysis*, *27*(2), 93–115. https://doi.org/10.1111/j.1538-4632.1995.tb00338.x

Apeagyei, A. E., Bisignano, C., Elliott, H., Hay, S. I., Lidral-Porter, B., Nam, S., Shyong, C., Tsakalos, G., Zlavog, B. S., Barış, E., Murray, C. J. L., & Dieleman, J. L. (2025). Tracking development

assistance for health, 1990–2030: Historical trends, recent cuts, and outlook. *The Lancet*, *406*(10501), 337–348. https://doi.org/10.1016/s0140-6736(25)01240-1

Bell, A., & Jones, K. (2015). Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data. *Political Science Research and Methods*, *3*(1), 133–153. https://doi.org/10.1017/psrm.2014.7

Bellemare, M. F., Masaki, T., & Pepinsky, T. B. (2017). Lagged explanatory variables and the estimation of causal effect. *The Journal of Politics*, *79*(3), 949–963. https://doi.org/10.1086/690946

Bendavid, E., & Bhattacharya, J. (2014). The relationship of health aid to population health improvements. *JAMA Internal Medicine*, *174*(6), 881–887. https://doi.org/10.1001/jamainternmed.2014.292

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Bergmeir, C., & Benítez, J. M. (2012). On the use of cross-validation for time series predictor evaluation. *Information Sciences*, *191*, 192–213. https://doi.org/10.1016/j.ins.2011.12.028

Bertram, M. Y., Lauer, J. A., De Joncheere, K., Edejer, T., Hutubessy, R., Kieny, M.-P., & Hill, S. (2016). Cost–effectiveness thresholds: Pros and cons. *Bulletin of the World Health Organization*, *94*(12), 925–930. https://doi.org/10.2471/blt.15.164418

Bhatt, S., Weiss, D. J., Cameron, E., Bisanzio, D., Mappin, B., Dalrymple, U., Battle, K., Moyes, C. L., Henry, A., Eckhoff, P. A., Wenger, E. A., Briët, O., Penny, M. A., Smith, T. A., Bennett, A., Yukich, J., Eisele, T. P., Griffin, J. T., Fergus, C. A., … Gething, P. W. (2015). The effect of malaria control on *Plasmodium falciparum* in Africa between 2000 and 2015. *Nature*, *526*(7572), 207–211. https://doi.org/10.1038/nature15535

Borlase, A., Le Rutte, E. A., Castaño, S., Blok, D. J., Toor, J., Giardina, F., Davis, E. L., & NTD Modelling Consortium. (2022). Evaluating and mitigating the potential indirect effect of COVID-19 on control programmes for seven neglected tropical diseases: A modelling study. *The Lancet Global Health*, *10*(11), e1600–e1611. https://doi.org/10.1016/S2214-109X(22)00360-6

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/a:1010933404324

Breusch, T. S., & Pagan, A. R. (1980). The Lagrange multiplier test and its applications to model specification in econometrics. *The Review of Economic Studies*, *47*(1), 239–253. https://doi.org/10.2307/2297111

Briggs, A. H., Weinstein, M. C., Fenwick, E. A. L., Karnon, J., Sculpher, M. J., & Paltiel, A. D. (2012). Model parameter estimation and uncertainty analysis: A report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group–6. *Medical Decision Making*, *32*(5), 722–732. https://doi.org/10.1177/0272989x12458348

Brunsdon, C., Fotheringham, A. S., & Charlton, M. E. (1996). Geographically weighted regression: A method for exploring spatial nonstationarity. *Geographical Analysis*, *28*(4), 281–298. https://doi.org/10.1111/j.1538-4632.1996.tb00936.x

Cavalcanti, D. M., de Oliveira Ferreira de Sales, L., da Silva, A. F., Basterra, E. L., Pena, D., Monti, C., Barreix, G., Silva, N. J., Vaz, P., Saute, F., Fanjul, G., Bassat, Q., Naniche, D., Macinko, J., & Rasella, D. (2025). Evaluating the impact of two decades of USAID interventions and projecting the effects of defunding on mortality up to 2030: A retrospective impact evaluation and forecasting analysis. *The Lancet*, *406*(10500), 283–294. https://doi.org/10.1016/s0140-6736(25)01186-9

Cohen, J. M., Smith, D. L., Cotter, C., Ward, A., Yamey, G., Sabot, O. J., & Moonen, B. (2012). Malaria resurgence: A systematic review and assessment of its causes. *Malaria Journal, 11*, Article 122. https://doi.org/10.1186/1475-2875-11-122

da Silva, A. F., Anderle, R. V. R., Barreix Sibils, G., de Oliveira Ferreira de Sales, L., Pena, D., Monti, C., Garcia Vaz, C., Santa-Ramírez, H.-A., Dos Santos de Jesus, G., Cavalcanti, D. M., Nhacolo, A., Macicame, I., Bassat, Q., & Rasella, D. (2026). Impact of two decades of humanitarian and development assistance and the projected mortality consequences of current defunding to 2030: Retrospective evaluation and forecasting analysis. *The Lancet Global Health*, *14*(5), e690–e701. https://doi.org/10.1016/S2214-109X(26)00008-2

da Silva, A. R., & Fotheringham, A. S. (2016). The multiple testing issue in geographically weighted regression. *Geographical Analysis*, *48*(3), 233–247. https://doi.org/10.1111/gean.12084

Dieleman, J. L., & Hanlon, M. (2014). Measuring the displacement and replacement of government health expenditure. *Health Economics*, *23*(2), 129–140. https://doi.org/10.1002/hec.3016

Driscoll, J. C., & Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, *80*(4), 549–560. https://doi.org/10.1162/003465398557825

Drukker, D. M. (2003). Testing for serial correlation in linear panel-data models. *The Stata Journal*, *3*(2), 168–177. https://doi.org/10.1177/1536867x0300300206

Elhorst, J. P. (2014). *Spatial econometrics: From cross-sectional data to spatial panels*. Springer. https://doi.org/10.1007/978-3-642-40340-8

Emadi, M., Delavari, S., & Bayati, M. (2021). Global socioeconomic inequality in the burden of communicable and non-communicable diseases and injuries: An analysis on Global Burden of Disease Study 2019. *BMC Public Health*, *21*(1), Article 1771. https://doi.org/10.1186/s12889-021-11793-7

Feng, L., Mo, G., Liu, Z., & Liu, Q. (2025). Global, regional, national burden, trends and health inequality of neglected tropical diseases and malaria from 1990 to 2021. *BMC Infectious Diseases*, *25*(1), Article 1649. https://doi.org/10.1186/s12879-025-12068-x

Fitzpatrick, C., Nwankwo, U., Lenk, E., de Vlas, S. J., & Bundy, D. A. P. (2017). An investment case for ending neglected tropical diseases. In K. K. Holmes, S. Bertozzi, B. R. Bloom, & P. Jha (Eds.), *Major infectious diseases* (3rd ed., Vol. 6, pp. 411–431). World Bank. https://doi.org/10.1596/978-1-4648-0524-0_ch17

Fotheringham, A. S., Yang, W., & Kang, W. (2017). Multiscale geographically weighted regression (MGWR). *Annals of the American Association of Geographers*, *107*(6), 1247–1265. https://doi.org/10.1080/24694452.2017.1352480

GBD 2019 Universal Health Coverage Collaborators. (2020). Measuring universal health coverage based on an index of effective coverage of health services in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019. *The Lancet*, *396*(10258), 1250–1284. https://doi.org/10.1016/s0140-6736(20)30750-9

GBD 2023 Causes of Death Collaborators. (2025). Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990–2023: A systematic analysis for the Global Burden of Disease Study 2023. *The Lancet*, *406*(10513), 1811–1872. https://doi.org/10.1016/S0140-6736(25)01917-8

GBD 2023 Disease and Injury and Risk Factor Collaborators. (2025). Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, including 660 subnational locations, 1990–2023: A systematic analysis for the

Global Burden of Disease Study 2023. *The Lancet*, *406*(10513), 1873–1922. https://doi.org/10.1016/s0140-6736(25)01637-x

Getis, A., & Aldstadt, J. (2004). Constructing the spatial weights matrix using a local statistic. *Geographical Analysis*, *36*(2), 90–104. https://doi.org/10.1111/j.1538-4632.2004.tb01127.x

Global Burden of Disease 2021 Health Financing Collaborator Network. (2023). Global investments in pandemic preparedness and COVID-19: Development assistance and domestic spending on health between 1990 and 2026. *The Lancet Global Health*, *11*(3), e385–e413. https://doi.org/10.1016/S2214-109X(23)00007-4

Greene, W. H. (2018). *Econometric analysis* (8th ed.). Pearson.

Grimes, J. E. T., Croll, D., Harrison, W. E., Utzinger, J., Freeman, M. C., & Templeton, M. R. (2014). The relationship between water, sanitation and schistosomiasis: A systematic review and meta-analysis. *PLOS Neglected Tropical Diseases*, *8*(12), Article e3296. https://doi.org/10.1371/journal.pntd.0003296

Hallett, T. B., Menzies, N. A., Resch, S., Pretorius, C., Stover, J., Su, J. S., Winskill, P., Gordon, M., Grahn, R., Mahmood, F., Smit, M., Hosseini, M., & Hunger, J. (2025). The case for optimal investment in combating HIV, tuberculosis, and malaria: A global modelling study. *The Lancet*, *406*(10500), 261–270. https://doi.org/10.1016/S0140-6736(25)00831-1

Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, *46*(6), 1251–1271. https://doi.org/10.2307/1913827

He, R., Zhu, W., Hui, S., Yu, M., Li, H., Li, Y., Huang, P., & Yu, R. (2025). Inequalities in disease burden and care quality of neglected tropical diseases and malaria, 1990–2021: Findings from the 2021 Global Burden of Disease Study. *PLOS ONE*, *20*(8), Article e0329475. https://doi.org/10.1371/journal.pone.0329475

Hoechle, D. (2007). Robust standard errors for panel regressions with cross-sectional dependence. *The Stata Journal*, *7*(3), 281–312. https://doi.org/10.1177/1536867x0700700301

Honaker, J., & King, G. (2010). What to do about missing values in time-series cross-section data. *American Journal of Political Science*, *54*(2), 561–581. https://doi.org/10.1111/j.1540-5907.2010.00447.x

Hsiao, A. J., & Emdin, C. A. (2014). The association between development assistance for health and malaria, HIV and tuberculosis mortality: A cross-national analysis. *Journal of Epidemiology and Global Health*, *5*(1), 41–48. https://doi.org/10.1016/j.jegh.2014.10.001

Huffstetler, H. E., Bandara, S., Bharali, I., Kennedy Mcdade, K., Mao, W., Guo, F., Zhang, J., Riviere, J., Becker, L., Mohamadi, M., Rice, R. L., King, Z., Farooqi, Z. W., Zhang, X., Yamey, G., & Ogbuoji, O. (2022). The impacts of donor transitions on health systems in middle-income countries: A scoping review. *Health Policy and Planning*, *37*(9), 1188–1202. https://doi.org/10.1093/heapol/czac063

Hurvich, C. M., Simonoff, J. S., & Tsai, C.-L. (1998). Smoothing parameter selection in nonparametric regression using an improved Akaike information criterion. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *60*(2), 271–293. https://doi.org/10.1111/1467-9868.00125

Imai, K., & Kim, I. S. (2021). On the use of two-way fixed effects regression models for causal inference with panel data. *Political Analysis*, *29*(3), 405–415. https://doi.org/10.1017/pan.2020.33

Jakubowski, A., Stearns, S. C., Kruk, M. E., Angeles, G., & Thirumurthy, H. (2017). The US President's Malaria Initiative and under-5 child mortality in sub-Saharan Africa: A difference-in-differences analysis. *PLOS Medicine, 14*(6), Article e1002319. https://doi.org/10.1371/journal.pmed.1002319

Kakwani, N., Wagstaff, A., & van Doorslaer, E. (1997). Socioeconomic inequalities in health: Measurement, computation, and statistical inference. *Journal of Econometrics, 77*(1), 87–103. https://doi.org/10.1016/s0304-4076(96)01807-6

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in neural information processing systems* (Vol. 30, pp. 3146–3154). https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

Khaki, J. J., Minnery, M., & Giorgi, E. (2025). Using ESPEN data for evidence-based control of neglected tropical diseases in sub-Saharan Africa: A comprehensive model-based geostatistical analysis of soil-transmitted helminths. *PLoS Neglected Tropical Diseases, 19*(1), Article e0012782. https://doi.org/10.1371/journal.pntd.0012782

Lee, L.-F., & Yu, J. (2010). Estimation of spatial autoregressive panel data models with fixed effects. *Journal of Econometrics, 154*(2), 165–185. https://doi.org/10.1016/j.jeconom.2009.08.001

LeSage, J., & Pace, R. K. (2009). *Introduction to spatial econometrics*. Chapman & Hall/CRC.

Liese, B. H., & Schubert, L. (2009). Official development assistance for health—how neglected are neglected tropical diseases? An analysis of health financing. *International Health, 1*(2), 141–147. https://doi.org/10.1016/j.inhe.2009.08.004

Lin, Y., Fang, K., Zheng, Y., Wang, H. L., & Wu, J. (2022). Global burden and trends of neglected tropical diseases from 1990 to 2019. *Journal of Travel Medicine, 29*(3), Article taac031. https://doi.org/10.1093/jtm/taac031

Lin, Y., Long, Q., Sun, J., Li, J., Li, Y., Liu, J., Zhang, Y., Wang, Y., & Jiang, S. (2025). Global trends, health inequalities, and projections in the burden of neglected tropical diseases and malaria from 1990 to 2021. *Scientific Reports, 15*, Article 20958. https://doi.org/10.1038/s41598-025-05530-y

Lo, N. C., Heft-Neal, S., Coulibaly, J. T., Leonard, L., Bendavid, E., & Addiss, D. G. (2019). State of deworming coverage and equity in low-income and middle-income countries using household health surveys: A spatiotemporal cross-sectional study. *The Lancet Global Health, 7*(11), e1511–e1520. https://doi.org/10.1016/S2214-109X(19)30413-9

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in neural information processing systems* (Vol. 30, pp. 4765–4774).

Mackenbach, J. P., & Kunst, A. E. (1997). Measuring the magnitude of socio-economic inequalities in health: An overview of available measures illustrated with two examples from Europe. *Social Science & Medicine, 44*(6), 757–771. https://doi.org/10.1016/S0277-9536(96)00073-1

Marseille, E., Larson, B., Kazi, D. S., Kahn, J. G., & Rosen, S. (2015). Thresholds for the cost–effectiveness of interventions: Alternative approaches. *Bulletin of the World Health Organization, 93*(2), 118–124. https://doi.org/10.2471/blt.14.138206

Mishra, P., & Newhouse, D. (2009). Does health aid matter? *Journal of Health Economics, 28*(4), 855–872. https://doi.org/10.1016/j.jhealeco.2009.05.004

Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika, 37*(1-2), 17–23. https://doi.org/10.1093/biomet/37.1-2.17

Morgenstern, H. (1995). Ecologic studies in epidemiology: Concepts, principles, and methods. *Annual Review of Public Health, 16*(1), 61–81. https://doi.org/10.1146/annurev.pu.16.050195.000425

Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica, 46*(1), 69–85. https://doi.org/10.2307/1913646

Murray, C. J. L., Ortblad, K. F., Guinovart, C., Lim, S. S., Wolock, T. M., Roberts, D. A., Dansereau, E. A., Graetz, N., Barber, R. M., Brown, J. C., Wang, H., Duber, H. C., Naghavi, M., Dicker, D., Dandona, L., Salomon, J. A., Heuton, K. R., Foreman, K., Phillips, D. E., … Vos, T. (2014). Global, regional, and national incidence and mortality for HIV, tuberculosis, and malaria during 1990–2013: A systematic analysis for the Global Burden of Disease Study 2013. *The Lancet, 384*(9947), 1005–1070. https://doi.org/10.1016/S0140-6736(14)60844-8

Namubiru, C. (2025). Institutional quality, aid flows, and malaria burden: A geospatial analysis of sub-Saharan Africa. *Malaria Journal, 24*(1), Article 332. https://doi.org/10.1186/s12936-025-05592-3

Natural Earth. (n.d.). *Admin 0 – Countries* [Map]. https://www.naturalearthdata.com/downloads/50m-cultural-vectors/50m-admin-0-countries-2/

O'Brien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & Quantity, 41*(5), 673–690. https://doi.org/10.1007/s11135-006-9018-6

O'Donnell, O., van Doorslaer, E., Wagstaff, A., & Lindelow, M. (2008). *Analyzing health equity using household survey data: A guide to techniques and their implementation*. World Bank.

Oshan, T. M., Li, Z., Kang, W., Wolf, L. J., & Fotheringham, A. S. (2019). mgwr: A Python implementation of multiscale geographically weighted regression for investigating process spatial heterogeneity and scale. *ISPRS International Journal of Geo-Information, 8*(6), Article 269. https://doi.org/10.3390/ijgi8060269

Patenaude, B. N. (2021). The relationship between development assistance for health and public health financing in 134 countries between 2000 and 2015. *Health Policy and Planning, 36*(4), 369–383. https://doi.org/10.1093/heapol/czab004

Pesaran, M. H. (2021). General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics, 60*(1), 13–50. https://doi.org/10.1007/s00181-020-01875-7

Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., & Pélissier, R. (2020). Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications, 11*(1), Article 4540. https://doi.org/10.1038/s41467-020-18321-y

Reeves, A., Gourtsoyannis, Y., Basu, S., McCoy, D., McKee, M., & Stuckler, D. (2015). Financing universal health coverage—Effects of alternative tax structures on public health systems: Cross-national modelling in 89 low-income and middle-income countries. *The Lancet, 386*(9990), 274–280. https://doi.org/10.1016/S0140-6736(15)60574-8

Rey, S. J., & Anselin, L. (2007). PySAL: A Python library of spatial analytical methods. *Review of Regional Studies, 37*(1), 5–27. https://doi.org/10.52324/001c.8285

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography, 40*(8), 913–929. https://doi.org/10.1111/ecog.02881

Rubin, D. B. (1987). *Multiple imputation for nonresponse in surveys*. Wiley.

Sempungu, J. K., Choi, M., Lee, E. H., & Lee, Y. H. (2023). The burden of malaria in sub-Saharan Africa and its association with development assistance for health and governance: An exploration of the

GBD 2019 study. *Global Public Health*, *18*(1), Article 2229892. https://doi.org/10.1080/17441692.2023.2229892

Shorrocks, A. F. (1980). The class of additively decomposable inequality measures. *Econometrica*, *48*(3), 613–625. https://doi.org/10.2307/1913126

Shroff, Z. C., Sparkes, S. P., Paina, L., Skarphedinsdottir, M., Gotsadze, G., Zakumumpa, H., Tang, K., Perera, P. N., Yuan, M., & Hanson, K. (2024). Managing transitions from external assistance: Cross-national learning about sustaining effective coverage. *Health Policy and Planning*, *39*(Supplement_1), i50–i64. https://doi.org/10.1093/heapol/czad101

Sterne, J. A. C., White, I. R., Carlin, J. B., Spratt, M., Royston, P., Kenward, M. G., Wood, A. M., & Carpenter, J. R. (2009). Multiple imputation for missing data in epidemiological and clinical research: Potential and pitfalls. *BMJ*, *338*, Article b2393. https://doi.org/10.1136/bmj.b2393

Stevens, G. A., Alkema, L., Black, R. E., Boerma, J. T., Collins, G. S., Ezzati, M., Grove, J. T., Hogan, D. R., Hogan, M. C., Horton, R., Lawn, J. E., Marušić, A., Mathers, C. D., Murray, C. J. L., Rudan, I., Salomon, J. A., Simpson, P. J., Vos, T., & Welch, V. (2016). Guidelines for accurate and transparent health estimates reporting: The GATHER statement. *The Lancet*, *388*(10062), e19–e23. https://doi.org/10.1016/s0140-6736(16)30388-9

Stolk, W. A., Kulik, M. C., le Rutte, E. A., Jacobson, J., Richardus, J. H., de Vlas, S. J., & Houweling, T. A. J. (2016). Between-country inequalities in the neglected tropical disease burden in 1990 and 2010, with projections for 2020. *PLoS Neglected Tropical Diseases*, *10*(5), Article e0004560. https://doi.org/10.1371/journal.pntd.0004560

Stover, J., Sonneveldt, E., Tam, Y., Horton, K. C., Phillips, A. N., Smith, J., Martin-Hughes, R., Ten Brink, D., Citron, D. T., Kim, H. Y., Akullian, A., Mudimu, E., Pickles, M., Bershteyn, A., Williamson, J., Meyer-Rath, G., Jamieson, L., Sully, E. A., White, J. N., … Kaftan, D. (2025). Effects of reductions in US foreign assistance on HIV, tuberculosis, family planning, and maternal and child health: A modelling study. *The Lancet Global Health*, *13*(10), e1669–e1680. https://doi.org/10.1016/S2214-109X(25)00281-5

Symons, T. L., Lubinda, J., McPhail, M., Saddler, A., van den Berg, M., Baggen, H., Berman, Y., Hafsia, S., Jayaseelen, R., Amratia, P., Browne, A., Cameron, E., Vargas-Ruiz, C., Rumisha, S. F., Golding, N., Weiss, D. J., & Gething, P. W. (2025). Estimating the potential malaria morbidity and mortality avertable by the US President's Malaria Initiative in 2025: A geospatial modelling analysis. *The Lancet*, *405*(10496), 2231–2240. https://doi.org/10.1016/S0140-6736(25)00805-0

Theil, H. (1967). *Economics and information theory*. North-Holland.

Tusting, L. S., Bottomley, C., Gibson, H., Kleinschmidt, I., Tatem, A. J., Lindsay, S. W., & Gething, P. W. (2017). Housing improvements and malaria risk in sub-Saharan Africa: A multi-country analysis of survey data. *PLOS Medicine*, *14*(2), Article e1002234. https://doi.org/10.1371/journal.pmed.1002234

Tusting, L. S., Willey, B., Lucas, H., Thompson, J., Kafy, H. T., Smith, R., & Lindsay, S. W. (2013). Socioeconomic development as an intervention against malaria: A systematic review and meta-analysis. *The Lancet*, *382*(9896), 963–972. https://doi.org/10.1016/S0140-6736(13)60851-X

United Nations Statistics Division. (n.d.). *Global SDG Indicators Database*. Retrieved October 7, 2026, from https://unstats.un.org/sdgs/dataportal

van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, *45*(3), 1–67. https://doi.org/10.18637/jss.v045.i03

von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Gøtzsche, P. C., & Vandenbroucke, J. P. (2007). The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE)

statement: Guidelines for reporting observational studies. *The Lancet*, *370*(9596), 1453–1457. https://doi.org/10.1016/s0140-6736(07)61602-x

Wagstaff, A., Paci, P., & van Doorslaer, E. (1991). On the measurement of inequalities in health. *Social Science & Medicine*, *33*(5), 545–557. https://doi.org/10.1016/0277-9536(91)90212-u

Weiss, D. J., Dzianach, P. A., Saddler, A., Lubinda, J., Browne, A., McPhail, M., Rumisha, S. F., Sanna, F., Gelaw, Y., Kiss, J. B., Hafsia, S., Jayaseelen, R., Baggen, H. S., Amratia, P., Bertozzi-Villa, A., Nesbit, O., Whisnant, J., Battle, K. E., Nguyen, M., … Gething, P. W. (2025). Mapping the global prevalence, incidence, and mortality of *Plasmodium falciparum* and *Plasmodium vivax* malaria, 2000–22: A spatial and temporal modelling study. *The Lancet*, *405*(10483), 979–990. https://doi.org/10.1016/S0140-6736(25)00038-8

White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, *30*(4), 377–399. https://doi.org/10.1002/sim.4067

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.

World Bank. (n.d.). *World Development Indicators*. Retrieved October 7, 2026, from https://databank.worldbank.org/source/world-development-indicators

World Health Organization. (n.d.). *Global Health Expenditure Database*. Retrieved October 7, 2026, from https://apps.who.int/nha/database

World Health Organization. (2020). *Ending the neglect to attain the Sustainable Development Goals: A road map for neglected tropical diseases 2021–2030*.

World Health Organization. (2025a). *Global report on neglected tropical diseases 2025*. https://www.who.int/publications/i/item/9789240114043

World Health Organization. (2025b). *World malaria report 2025: Addressing the threat of antimalarial drug resistance*. https://www.who.int/publications/i/item/9789240117822

World Health Organization Regional Office for Africa. (2025). *Sustaining momentum in NTD programmes amidst a reduced funding landscape: A regional assessment*. https://espen.afro.who.int/sites/default/files/content/document/Sustaining%20Momentum%20in%20NTD%20Programmes%20Amidst%20a%20Reduced%20Funding%20Landscape_A%20Regional%20Assessment_0.pdf

Yan, I., Korenromp, E., & Bendavid, E. (2015). Mortality changes after grants from the Global Fund to Fight AIDS, Tuberculosis and Malaria: An econometric analysis from 1995 to 2010. *BMC Public Health*, *15*, Article 977. https://doi.org/10.1186/s12889-015-2305-1

Zheng, J.-X., Liu, Y., Guan, S.-Y., Guo, Z.-Y., Duan, L., Lv, S., Zhang, S.-X., Li, S.-Z., Zhou, X.-N., & Feng, X.-Y. (2025). Global, regional, and national burden of neglected tropical diseases and malaria in the general population, 1990–2021: Systematic analysis of the Global Burden of Disease Study 2021. *Journal of Advanced Research*, *79*, 769–781. https://doi.org/10.1016/j.jare.2025.04.004

Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *67*(2), 301–320. https://doi.org/10.1111/j.1467-9868.2005.00503.x

Tables

Table 1: Burden of Neglected Tropical Diseases and Malaria by GBD Super-Region, 2023

Super-region	Countries	Population (millions)	NTD DALYs, millions (% of global)	NTD age-standardised rate	Malaria DALYs, millions (% of global)	Malaria age-standardised rate	% change in age-standardised NTD + malaria DALY rate, 2010–2023 [95% UI]
Sub-Saharan Africa	46	1,208.6	7.75 (40.3)	644.6	49.03 (95.1)	3,237.1	-33.8 [-49.9, -21.1]
South Asia	5	1,887.5	4.82 (25.1)	263.9	1.71 (3.3)	95.7	-43.8 [-52.9, -29.2]
Southeast Asia, East Asia, and Oceania	34	2,158.3	3.50 (18.2)	179.4	0.28 (0.5)	15.9	-38.2 [-56.3, -7.1]
Latin America and Caribbean	33	588.6	1.55 (8.1)	256.8	0.07 (0.1)	12.6	-22.7 [-30.8, -12.9]
North Africa and Middle East	21	674.1	1.02 (5.3)	165.2	0.48 (0.9)	75.5	-19.8 [-29.2, -6.5]
Central Europe, Eastern Europe, and Central Asia	29	409.2	0.25 (1.3)	55.2	0.00 (0.0)	0.0	-13.0 [-32.3, 14.3]
High-income	36	1,096.4	0.34 (1.8)	26.5	0.00 (0.0)	0.0	-7.5 [-26.3, 11.0]
Global	**204**	**8,022.6**	**19.24 (100)**	**—**	**51.56 (100)**	**—**	**-21.6 [-36.3, -7.8]**

Note. NTD = neglected tropical diseases (19 GBD causes, excluding malaria); DALY = disability-adjusted life-year; UI = uncertainty interval. Rates are age-standardised DALYs per 100 000 population. Percentage changes are GBD 2023 published estimates; UIs are the 2.5th–97.5th percentiles of draw-level changes. Population from World Bank WDI (latest year). Source: GBD 2023 (IHME).

Table 2: Spatial Autocorrelation and Socioeconomic Inequality in Age-Standardised DALY Rates, 2023

Outcome	Global Moran's I	*z*	*p*	High–High clusters (n)	Low–Low clusters (n)	Spatial outliers (n)	Concentration index by SDI [95% CI]	Theil T	Between / within super-region (%)
NTDs (excl. malaria)	0.71	19.26	< .001	48	39	0	-0.43 [-0.54, -0.27]	0.43	63.4 / 36.6
Malaria	0.79	21.11	< .001	42	62	1	-0.80 [-0.88, -0.65]	1.34	78.9 / 21.1

Note. Moran's I and local indicators of spatial association (LISA) computed on log(1 + rate) with row-standardised k = 6 nearest-neighbour weights and 9 999 permutations; LISA clusters reported at Benjamini–Hochberg false-discovery-rate q < .05 (n = 203 countries with geometry). Concentration index is population-weighted, ranking countries by Socio-demographic Index (SDI); negative values indicate burden concentrated in low-SDI countries; 95% CI from 2 000 bootstrap resamples. Theil T is population-weighted and decomposed into between- and within-super-region components. The reported Theil estimates exclude zero rates and require population data (NTDs: n = 199; malaria: n = 86); ~~Table S15 gives~~Table S22 gives the zero-inclusive sensitivity.

Table 3: Association Between External (Donor) Share of Health Expenditure and Age-Standardised DALY Rates Across Estimators

Estimator	NTDs: % change [95% CI]	*p*	*n*	Malaria: % change [95% CI]	*p*	*n*
Between-country (2023 cross-section)						
OLS + super-region FE	-13.43 [-19.97, -6.36]	< .001	145	-19.90 [-41.07, 8.87]	.156	58
Spatial error model	-11.17 [-18.32, -3.38]	.006	145	-9.75 [-33.19, 21.90]	.503	58
Within-country panel (2007–22)						
Pooled OLS	-4.46 [-9.54, 0.91]	.102	2,287	16.62 [-7.05, 46.32]	.184	1,187
Random effects	-1.04 [-3.14, 1.10]	.339	2,287	-6.04 [-17.57, 7.10]	.350	1,187
Two-way FE (cluster)	-0.64 [-2.49, 1.24]	.503	2,287	-6.32 [-17.14, 5.91]	.297	1,187
Two-way FE, Driscoll–Kraay SE (preferred)	**-0.64 [-2.39, 1.14]**	**.478**	**2,287**	**-6.32 [-10.85, -1.56]**	**.010**	**1,187**
First difference	-1.48 [-3.16, 0.23]	.089	2,134	-1.95 [-4.96, 1.15]	.215	1,110
FE + country trends	0.17 [-1.31, 1.67]	.825	2,287	-2.06 [-8.10, 4.38]	.522	1,187
Mundlak CRE	-1.14 [-3.31, 1.08]	.313	2,287	-6.41 [-18.40, 7.35]	.344	1,187
Spatial panel 2007–2022 — provisional						
Spatial lag FE (SAR)	-0.45 [-1.47, 0.58]	.386	2,080	-2.79 [-8.41, 3.17]	.351	1,136
Spatial error FE (SEM)	-0.20 [-1.21, 0.82]	.704	2,080	-2.77 [-8.34, 3.13]	.350	1,136

Note. Estimates are exponentiated coefficients per 10-percentage-point increase in external expenditure share. For NTDs, they represent percentage differences in age-standardised DALY rates; for malaria, the transformation applies to rate + 1. Models adjust for log LDI, UHC, sanitation and drinking-water coverage, with rural population share in cross-sectional models and lagged domestic government spending in panel models. Panel financing exposures are lagged 2 years. The financing sample excludes the GBD high-income super-region; malaria models are restricted to endemic locations. n denotes locations in cross-sections and country-years in panels. FE = fixed effects; CRE = correlated random effects; SAR = spatial autoregressive model; SEM = spatial error model. Spatial models use balanced panels and six-nearest-neighbour weights; SAR entries are exponentiated coefficients, with direct, indirect and total associations in Table S12. Diagnostics are reported in Table S3. Spatial-panel rows remain provisional because the main and supplementary tables read different model outputs.

Table 4: ~~Provisional annual DALY projections under development-assistance cuts~~Projected Excess Annual DALYs Under Development-Assistance Cut Scenarios, Low- and Middle-Income Countries

Outcome	Accounting model: excess DALYs, thousands [95% UI]	% of 2023 burden	Sub-Saharan Africa share (%)	Econometric model: excess DALYs, thousands [95% UI]	Economic value, US$ billions [95% UI]
Mild (5–15% cut)					
NTDs (excl. malaria)	~~108 [55, 171]~~182 [94, 277]	~~0.6~~1.0	~~93~~91	10 [-20, 43]	~~0.14 [0.07, 0.24]~~0.25 [0.13, 0.39]
Malaria	825 [434, 1,244]	1.6	96	~~489 [107, 1,116]~~490 [107, 1,116]	0.94 [0.49, 1.44]
Reference (19–26% cut)					
NTDs (excl. malaria)	~~241 [192, 318]~~407 [335, 502]	~~1.3~~2.2	~~94~~91	25 [-42, 90]	~~0.32 [0.24, 0.47]~~0.55 [0.45, 0.72]
Malaria	1,850 [1,514, 2,205]	3.6	96	~~1,222 [270, 2,287]~~1,222 [270, 2,288]	2.11 [1.65, 2.55]
Severe (40–60% cut)					
NTDs (excl. malaria)	~~540 [404, 718]~~909 [708, 1,140]	~~2.9~~4.8	~~93~~91	56 [-101, 227]	~~0.71 [0.52, 1.03]~~1.24 [0.95, 1.62]
Malaria	4,099 [3,255, 5,068]	8.0	96	~~3,059 [701, 5,970]~~3,060 [701, 5,972]	4.67 [3.60, 5.85]

Note. The accounting model multiplies the positive reduction in all-age DALY rate from 2010 to 2023 by 2023 population / 100 000, external expenditure share and the proportional cut. The econometric model applies the preferred panel coefficient to the cut-induced change in external expenditure share. Intervals derive from 4000 simulations under the stated burden, correlation, cut and coefficient assumptions. Monetary values use one times GDP per capita per DALY; 0.5 and 3 times are examined in Table S7. Estimates are conditional annual burden projections, not observed or first-year effects. ~~All values are retained original outputs, not validated revised projections. Missing NTD baselines and the malaria outcome transformation require correction before finalisation.~~NTD baselines cover all 19 causes; malaria projections are computed on the (1 + rate) scale of the fitted model.